

13

MEASUREMENT AND SPACE

CONGRUENT AND SIMILAR FIGURES

Unique to Australia are a number of tourist attractions known as the 'Bigs' or 'Biggies'.

- the Big Merino (Goulburn, NSW)
- the Big Banana (Coffs Harbour, NSW)
- the Big Golden Guitar (Tamworth, NSW)
- the Big Pineapple (Woombye, Qld)
- the Big Orange (Berri, SA)
- the Big Crocodile (Wyndham, WA)
- the Big Captain Cook (Cairns, Qld)
- the Big Mango (Bowen, Qld)
- the Big Rocking Horse (Gumeracha, SA)
- the Giant Koala (Dadsells Bridge, Vic)

Each attraction is a scale model or enlargement of an Australian object.
The Big Pineapple is 16 metres high.



Chapter outline

	Working mathematically				
13.01 Congruent figures*	U	F		R	C
13.02 Tests for congruent triangles*	U	F		R	C
13.03 Congruent triangle proofs*	U	F		R	C
13.04 Using congruence to prove geometrical properties*	U	F		R	C
13.05 Similar figures	U	F		R	C
13.06 Finding sides in similar figures	U	F	PS	R	
13.07 Scale drawings	U	F	PS	R	
13.08 Areas of similar figures*	U	F		R	
13.09 Tests for similar triangles*	U	F		R	C
13.10 Similar triangles proofs*	U	F		R	C

***PATH**

Wordbank

congruence test One of 4 tests for proving that triangles are congruent: SSS, SAS, AAS and RHS

congruent Identical; exactly the same (symbol: ≡)

enlargement An increase in the size of a shape

image A transformed shape after it has been enlarged or reduced

included angle The angle between two given sides of a shape

scale factor The amount by which a shape has been enlarged or reduced, equal to $\frac{\text{image length}}{\text{original length}}$

similar To have the same shape but not necessarily the same size, an enlargement or reduction (symbol: ~)

superimpose To place one figure on top of another figure so that sides and angles match

Quiz
Wordbank 13

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (9):

13.02 Test for congruent triangles
13.03 Congruent triangles proofs
13.04 Congruent triangles proofs • Proving properties of a rectangle
13.06 Finding an unknown side in similar figures
13.07 Scale drawings
13.08 Areas of similar figures
13.09 Similar triangles • Test for similar triangles

Twig videos (5):

13.07 Jai Singh (sundial) • Modelling the Spitfire • Queen Hatshepsut's ship • The history of the golden ratio • Fractals: The Koch snowflake

Quizzes (5):

- Wordbank 13
- SkillCheck 13
- Mental skills 13
- Language of maths 13
- Test yourself 13

Skillsheet (1):

13.06 Finding sides in similar triangles

Worksheets (16):

13.01 Congruent or different triangles? • Drawing different triangles • Triangle constructions group clues
13.04 Proving properties of quadrilaterals
13.05 A page of similar figures • A page of congruent and similar figures • Enlargements and reductions • Enlarging a logo • 5 mm grid paper
13.06 Finding sides in similar figures • Congruence and similarity review
13.07 Problems involving scale drawings
13.10 Congruent and similar triangle proofs • Complete the proofs • Investigating paper sizes
Mind map Congruent and similar figures
Test yourself 5 5 mm grid paper

Puzzles (4):

13.05 Cartoon enlargement
13.06 Similar triangles • Nested similar triangles
Language of maths Congruence and similarity crossword

Technology (2):

13.07 Converting map scales to ratios • Area of similar shapes

Spreadsheets (2):

13.07 Converting map scales to ratios • Area of similar shapes

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- (PATH) identify congruent figures and investigate their properties
- (PATH) investigate the 4 tests for congruent triangles and write formal proofs for congruent triangles
- (PATH) investigate properties of quadrilaterals using congruent triangles and solve related geometry problems
- identify similar figures and investigate their properties
- use scale factors to find unknown sides in similar figures
- measure and calculate using scale drawings
- (PATH) investigate ratios of areas of similar figures
- (PATH) investigate the 4 tests for similar triangles and write formal proofs for similar triangles

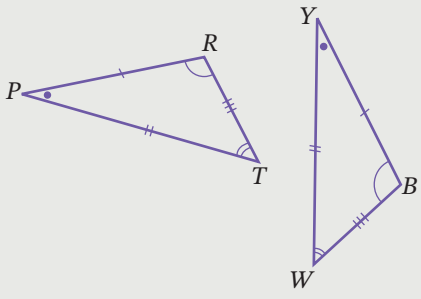
SkillCheck ANSWERS ON P. 663

Quiz
SkillCheck 13

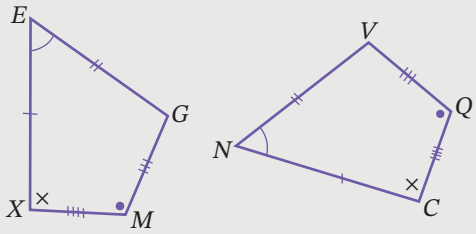
1 For each pair of identical figures, list all pairs of:

i matching angles **ii** matching sides

a



b



2 Solve each equation.

a $\frac{x}{24} = \frac{5}{8}$ **b** $\frac{m}{7} = \frac{10}{14}$ **c** $\frac{a}{2} = \frac{4}{8}$ **d** $\frac{k}{15} = \frac{2}{3}$

e $\frac{7}{12} = \frac{x}{3}$ **f** $\frac{5}{6} = \frac{n}{4}$ **g** $\frac{t}{2} = \frac{7}{3}$ **h** $\frac{d}{5} = \frac{17}{10}$

13.01

Congruent figures

13.01

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Congruent figures

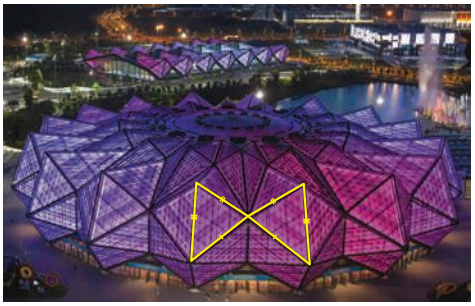
2 figures are **congruent** if they are identical in shape and size. For **congruent figures**:

- **matching sides** are equal
- **matching angles** are equal

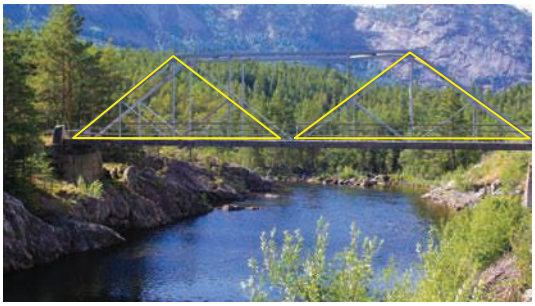
Matching sides are **corresponding sides** that are in the same position.

Matching angles are **corresponding angles** that are in the same position.

One way of testing whether 2 figures are congruent is to **superimpose** one shape onto the other, that is, to move it to a position on top of the other so that sides and angles match. Congruent figures may be identified by superimposing them through a combination of translations (slides), rotations (turns) and reflections (flips).



Alamy Stock Photo/View Stock



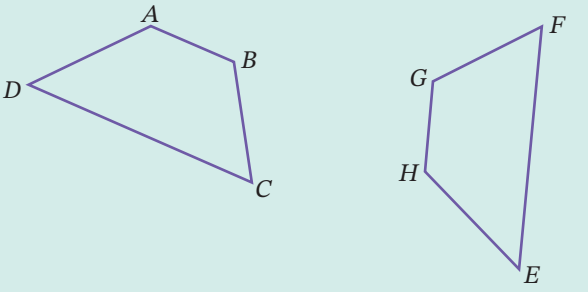
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Congruent triangles are commonly seen in the construction of bridges and buildings because they are considered more stable and strong to use.

Example 1

The 2 quadrilaterals *ABCD* and *EFGH* are congruent. List all pairs of matching sides and matching angles.



Solution

By rotating the figure *ABCD* anticlockwise, its shape can be superimposed exactly on *EFGH*.

The pairs of matching sides are:

AB and *HG*

BC and *GF*

CD and *FE*

AD and *HE*

The pairs of matching angles are:

$\angle A$ and $\angle H$

$\angle B$ and $\angle G$

$\angle C$ and $\angle F$

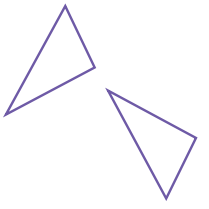
$\angle D$ and $\angle E$

EXERCISE 13.01 ANSWERS ON P. 663

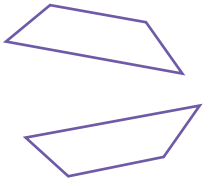
Congruent figures U F R C

1 Is each pair of figures congruent? Why? R C

a



b



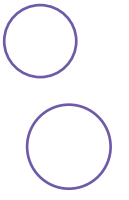
c



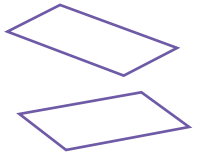
d



e



f



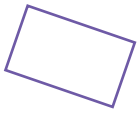
Foundation Standard Complex

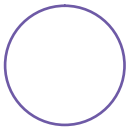
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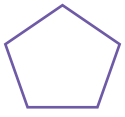
PATH

13.01

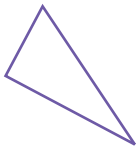
2 Match shapes that are congruent.

a 

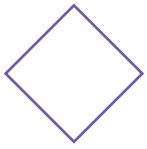
b 

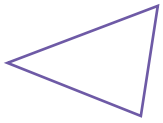
c 

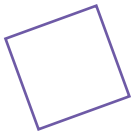
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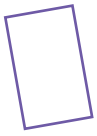
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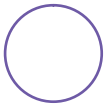
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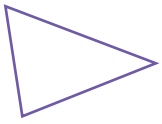
g 

h 

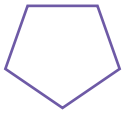
i 

j 

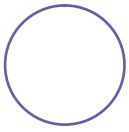
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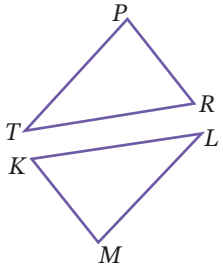
o 

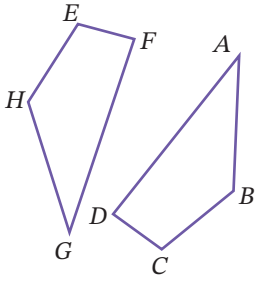
p 

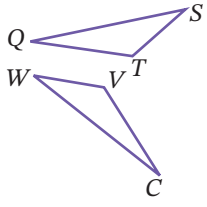
3 For each pair of congruent figures, list all pairs of:

i matching sides **ii** matching angles

EXAMPLE 1

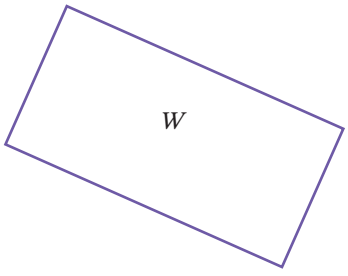
a 

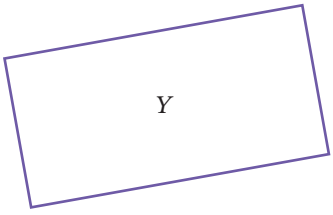
b 

c 

4 **a** For each rectangle, measure:

i its perimeter **ii** its area





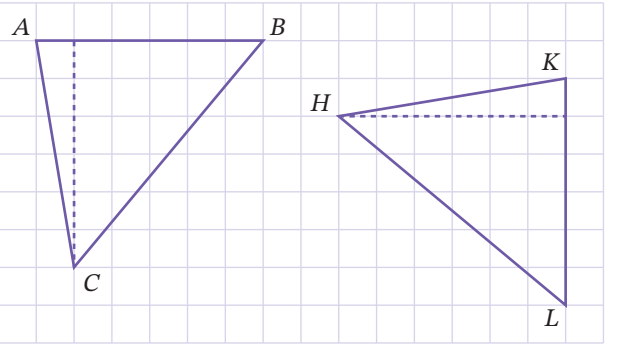
b Are the 2 rectangles congruent?

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5 **a** For each triangle (drawn on 5 mm grid), measure:

i its perimeter **ii** its area.



b Are the triangles congruent?

6 Copy and complete:

The p_____ and a_____ of congruent figures are e_____.

7 Draw 2 triangles that have the same perimeter but are not congruent. R C

8 Draw 2 triangles that have the same area but are not congruent. R C

9 Write True or False for each statement. R C

a If the perimeters of 2 rectangles are equal, then the 2 rectangles must be congruent.

b If the areas of 2 rectangles are equal, then the 2 rectangles must be congruent.

c If the perimeters and areas of 2 rectangles are equal, then the 2 rectangles must be congruent.

d Congruent figures must have the same perimeters and areas.

Technology

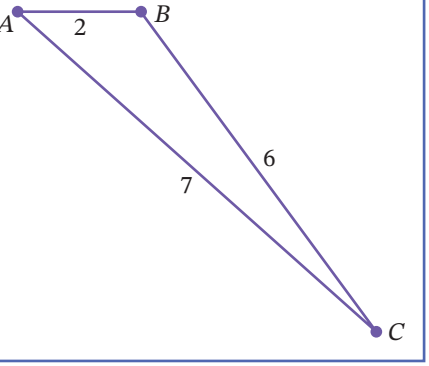
Tests for congruent triangles

Use dynamic geometry software to test whether 2 triangles are congruent.
It is best to do this activity in small groups.

Given 3 sides: SSS

1 Construct a triangle with side lengths 2 cm, 6 cm and 7 cm.

2 Can you construct a **different** triangle with the same 3 side lengths? What do you notice about the second triangle? Is it congruent (exactly the same shape and size) to the first triangle?



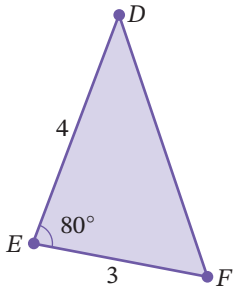
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Given 2 sides and the included angle: SAS

1 Construct a triangle with sides of length 3 cm and 4 cm, and the angle between them to be exactly 80°.

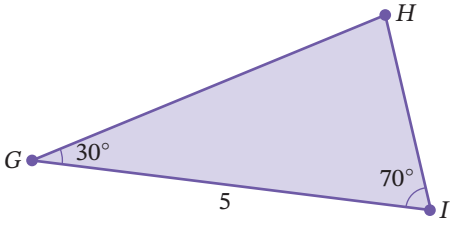
2 Can you construct a different triangle with the same 2 sides and included angle of 80°? Or is it congruent to the first triangle?



Given 2 angles and a side: AAS

1 Construct a triangle with a side length of 5 units between 2 angles of size 30° and 70°.

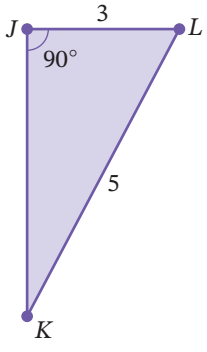
2 Can you construct a different triangle with the same 2 angles and a matching side of 5 cm? Or is it congruent to the first triangle?



Given a right angle, hypotenuse and side: RHS

1 Construct a right-angled triangle with one of the shorter sides being 3 cm and the hypotenuse 5 cm.

2 Can you construct a different triangle with the same 2 sides and right angle? Or is it congruent to the first triangle?

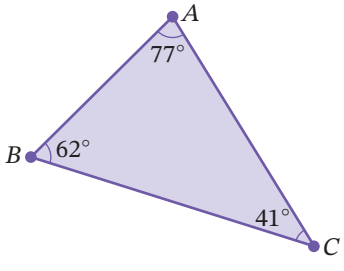


Given 3 angles: AAA

1 Construct a triangle with angles 77°, 62° and 41°.

2 Can you construct a different triangle with the 3 angles? Or is it congruent to the first triangle?

3 Are all triangles with angles 77°, 62° and 41° congruent?



PATH

Worksheets
Congruent or different triangles?

Drawing different triangles

Triangle constructions group clues

Investigation

Congruent triangles

Work in groups of 3 or 4 to complete this investigation.

You will need: paper, a ruler, a pencil, a protractor, compasses and scissors.

1 **a** Individually, construct a triangle with side lengths 5 cm, 7 cm and 9 cm.

b Use your protractor to measure the angles of the triangle.

c Compare your results with those of the students in your group. Are all your group's triangles congruent?

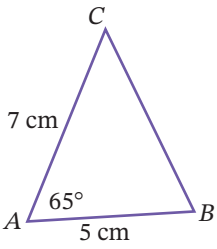
d Use your results to decide whether each statement is true or false.

i *If 3 sides of one triangle are respectively equal to the 3 sides of another triangle, then the matching angles of the triangles are equal.*

ii *If all sides of one triangle are equal to all sides of another triangle, then the 2 triangles are congruent.*

2 **a** Construct $\triangle ABC$ so that $AB = 5$ cm, $AC = 7$ cm and $\angle A = 65°$.

Since $\angle A$ lies between AB and AC, $\angle A$ is called the **included angle**.



b Measure the third side of the triangle, and the sizes of the other two angles.

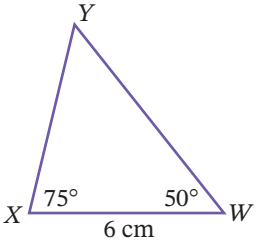
c Compare your results with those of others in your group.

d Use your results to decide whether this statement is true or false:

If two sides and the included angle of one triangle are respectively equal to 2 sides and the included angle of another triangle, then the 2 triangles are congruent.

3 **a** Construct $\triangle XWY$ with $XW = 6$ cm, $\angle X = 75°$ and $\angle W = 50°$.

b Measure $\angle Y$ and the other 2 sides of the triangle.



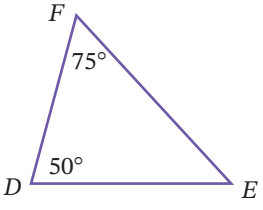
c Compare your results with those of your group members.

d Now construct $\triangle DEF$ with $DE = 6$ cm, $\angle D = 50°$ and $\angle F = 75°$.

e Are $\triangle XWY$ and $\triangle DEF$ congruent? Give reasons for your answer.

f Use your results to decide whether this statement is true:

If 2 angles and one side of one triangle are respectively equal to 2 angles and the matching side of another triangle, then the 2 triangles are congruent.



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4 a Construct $\triangle GHK$ with $\angle K = 90°$, $GH = 8$ cm, $GK = 5$ cm.

b What is another name for the side GH?

c Measure the other side and angles of the triangle and compare your results with those of other members in your group. Are their triangles congruent to your triangle?

d Use your results to decide whether the following statement is true or false:

If the hypotenuse and a second side of one right-angled triangle are respectively equal to the hypotenuse and a second side of another right-angled triangle, then the triangles are congruent.

5 a Draw a triangle with angles 40°, 63° and 77°.

b Compare your triangle with those of other members of your group. Are the triangles congruent? Give reasons.

6 Use your results to determine what information is needed to prove that 2 triangles are congruent.

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Tests for congruent triangles 13.02

PATH

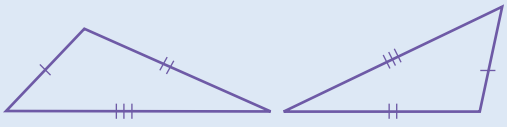
There are 4 sets of conditions that can be used to determine if 2 triangles are congruent. These are called the **tests for congruent triangles** or **congruence tests**.

Congruence tests

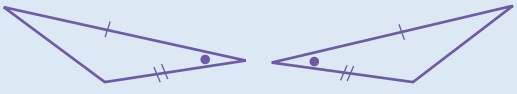
There are 4 tests for congruent triangles: **SSS**, **SAS**, **AAS** or **RHS**.

Two triangles are congruent if:

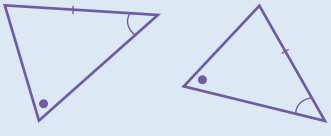
- the 3 sides of one triangle are respectively equal to the three sides of the other triangle **(SSS rule)**



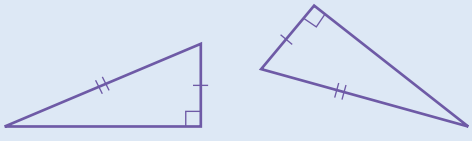
- 2 sides and the **included angle** of one triangle are respectively equal to 2 sides and the **included** angle of the other triangle **(SAS rule)**



- 2 angles and one side of one triangle are respectively equal to 2 angles and the matching side of the other triangle **(AAS rule)**



- they are right-angled and the hypotenuse and another side of one triangle are respectively equal to the hypotenuse and another side of the other triangle **(RHS rule).**

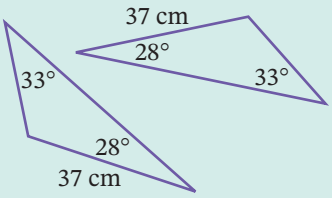


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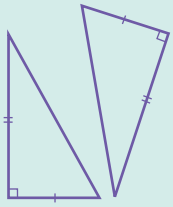
Example 2

Which congruence test (SSS, SAS, AAS or RHS) can be used to prove that each pair of triangles are congruent?

a



b



Solution

a 2 angles and one side of one triangle are equal to 2 matching angles and one matching side of the other triangle.

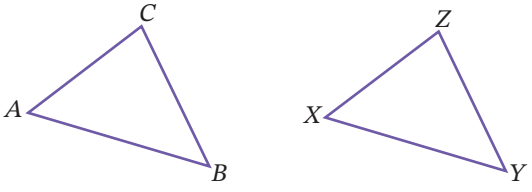
$\therefore$ The congruence test is AAS.

b 2 sides and the included angle of one triangle are equal to 2 matching sides and the included angle of the other triangle.

$\therefore$ The congruence test is SAS.

The congruence symbol $\equiv$

The symbol for 'is congruent to' is a special equals sign, written as '$\equiv$' (which also means 'is identical to'). The 2 triangles below are congruent, so we can write $\triangle \boldsymbol{ABC} \equiv \triangle \boldsymbol{XYZ}$, which is read: 'triangle *ABC* is congruent to triangle *XYZ*'.



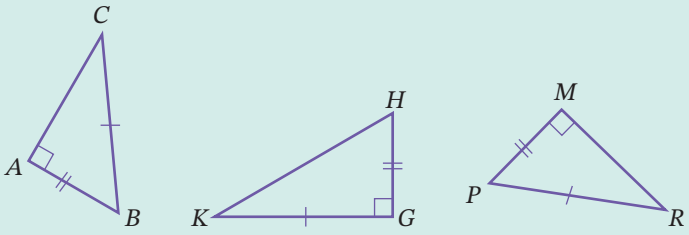
When using this notation, we must make sure that the vertices (and angles) of the congruent figures are written in matching order:

$\triangle ABC \equiv \triangle XYZ$ means $\angle A = \angle X$, $\angle B = \angle Y$, $\angle C = \angle Z$

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Example 3

a Decide which 2 of these triangles are congruent and state the congruence test used.



b Use the correct notation to write a congruency statement relating the 2 triangles.

Solution

a The 3 right-angled triangles all have 2 equal sides, but BC and PR are both hypotenuses while KG is not.

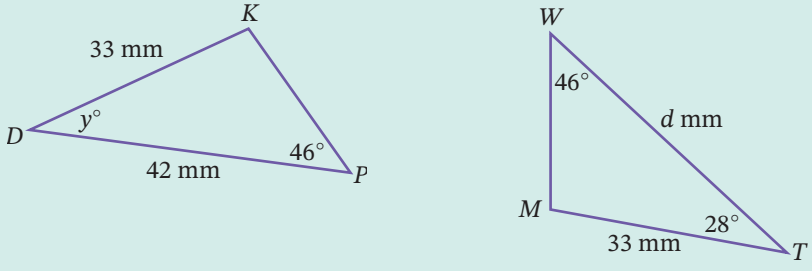
$\therefore \triangle CAB$ and $\triangle MPR$ are congruent by the RHS rule.

b $\angle A$ matches with $\angle M$, $\angle B$ matches with $\angle P$, $\angle C$ matches with $\angle R$. Matching order of vertices

$\therefore \triangle ABC \equiv \triangle MPR$

Example 4

If the 2 triangles are congruent, find the value of each variable.



Solution

Since $\triangle DPK \equiv \triangle TWM$,

$DP = TW$ Matching sides of congruent triangles

$\therefore d = 42$

$\angle T = \angle D$ Matching angles of congruent triangles

$\therefore y = 28$

PATH

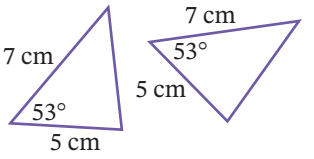
EXERCISE 13.02 ANSWERS ON P. 663

Tests for congruent triangles U F R C

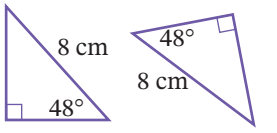
EXAMPLE 2

1 Which congruence test (SSS, SAS, AAS or RHS) can be used to prove that each pair of triangles are congruent? R C

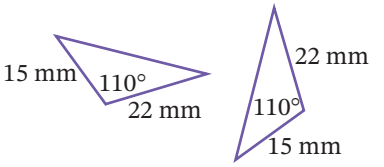
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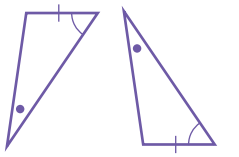
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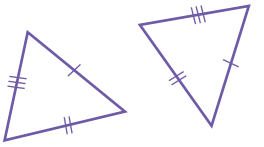
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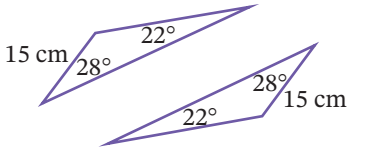
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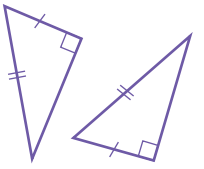
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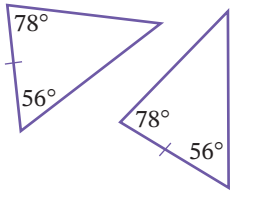
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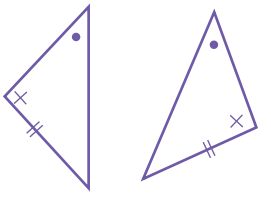
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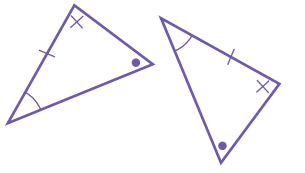
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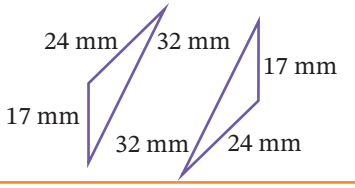
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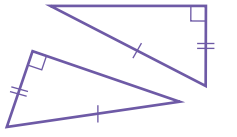
j



k



l



Foundation Standard Complex

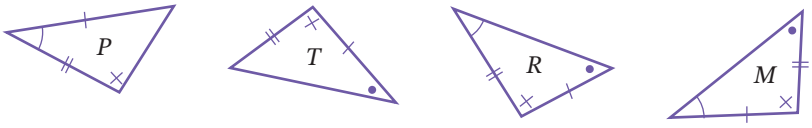
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PATH

13.02

2 Find the 2 congruent triangles in each set. Select the correct answer **A**, **B**, **C** or **D**.

a



A P and T **B** T and R **C** M and R **D** M and P

b



A W and Y **B** Y and X **C** X and V **D** W and V

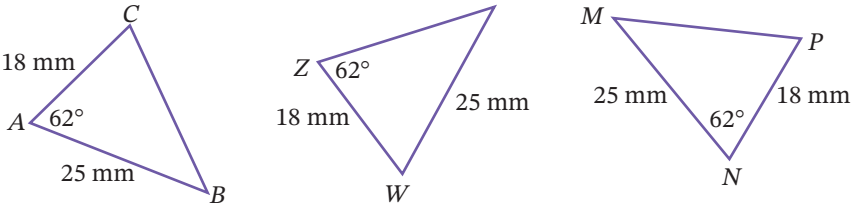
EXAMPLE 3

3 For each set of triangles: R C

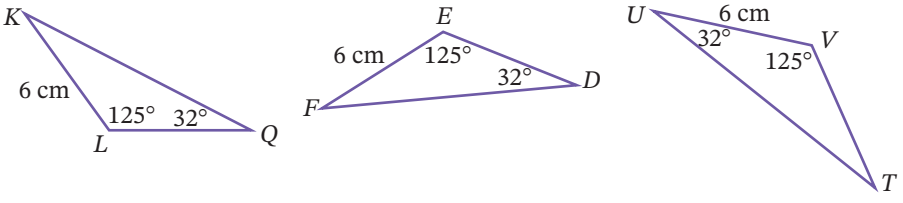
i decide which 2 are congruent and state the congruence test used

ii use the correct notation to write a congruency statement relating those 2 triangles.

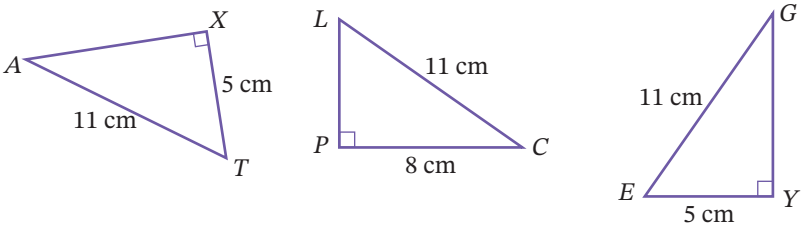
a



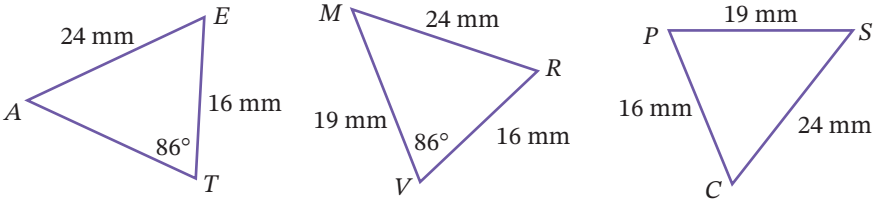
b



c



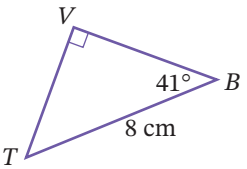
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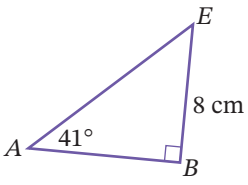


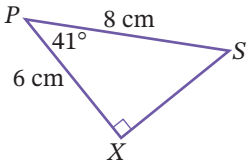
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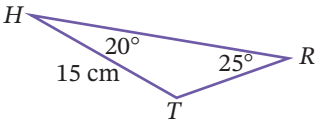
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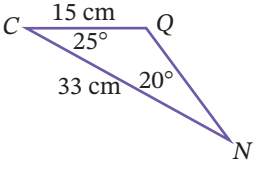


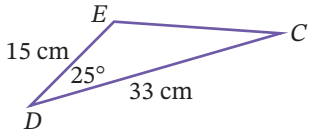




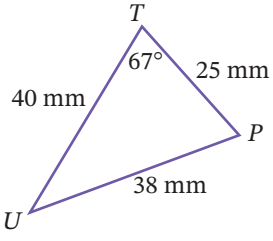
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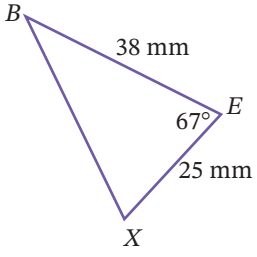


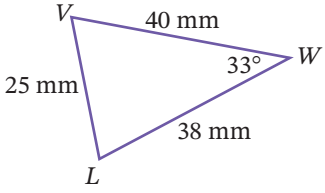




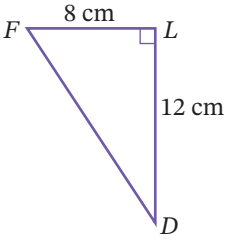
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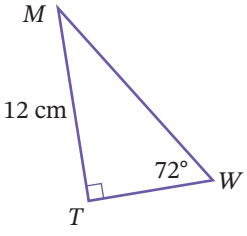


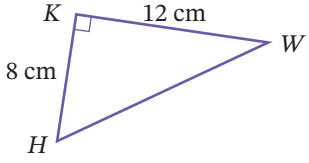




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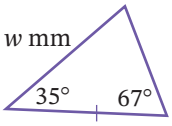


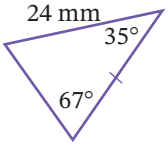


EXAMPLE 4

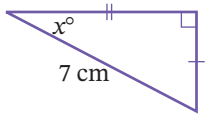
4 Find the value of the variable in each pair of congruent triangles. R

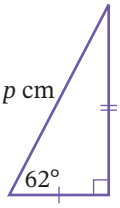
a



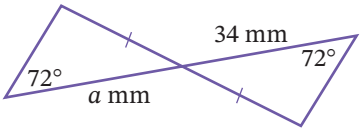


b

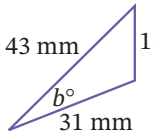


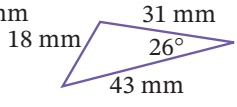


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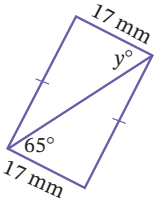


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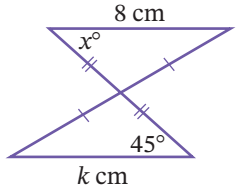




e



f



Foundation Standard Complex

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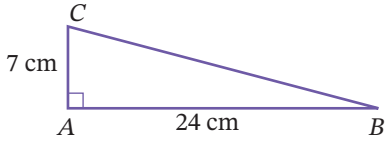
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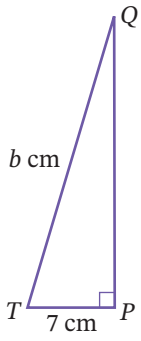
13.02

5 $\triangle ABC \equiv \triangle TPQ$. R

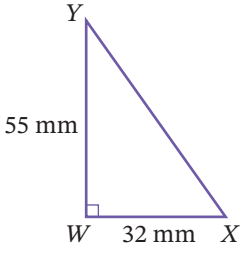
a Use Pythagoras' theorem to find BC.

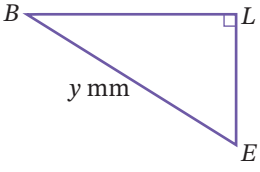
b Find the value of b.



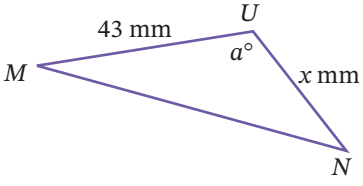


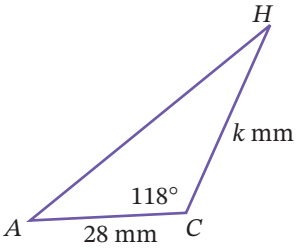
6 If $\triangle YXW \equiv \triangle BEL$, find y correct to one decimal place. R





7 If $\triangle NUM \equiv \triangle ACH$, find the value of each variable. R





8 The diagram shows 4 right-angled triangles joined together. State whether each pair of triangles are congruent, and if so, state the test used. R C

a W and X

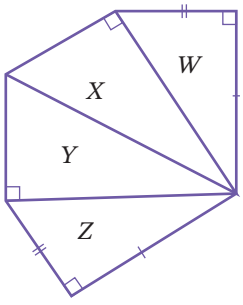
b W and Y

c W and Z

d X and Y

e X and Z

f Y and Z



Foundation Standard Complex

13.03 Congruent triangle proofs

PATH

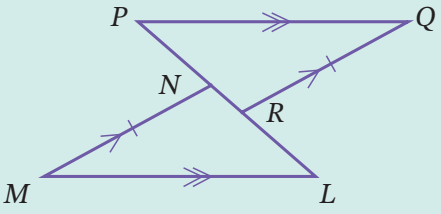
Video
Congruent triangle proofs

To formally prove that 2 triangles are congruent, there is a specific format using one of the 4 congruence tests SSS, SAS, AAS or RHS.

Example 5

In the diagram, $PQ \parallel LM$, $QR \parallel MN$ and $QR = MN$.

Prove that $\triangle PQR \equiv \triangle LMN$.



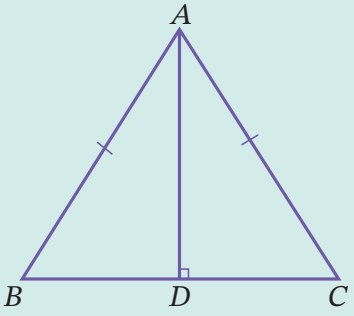
Solution

In $\triangle PQR$ and $\triangle LMN$:	Identifying the triangles in matching order of vertices
$QR = MN$ (given)	Stating each part of the congruence test, giving reasons
$\angle P = \angle L$ (alternate angles, $PQ \parallel LM$)	
$\angle QRP = \angle MNL$ (alternate angles, $QR \parallel MN$)	
$\therefore \triangle PQR \equiv \triangle LMN$ (AAS)	Concluding the congruence proof, stating the test used

Example 6

In $\triangle ABC$, $AB = AC$ and $AD \perp BC$.

Prove that $\triangle ABD \equiv \triangle ACD$ and hence AD bisects $\angle BAC$.



Solution

In $\triangle ABD$ and $\triangle ACD$:

$AB = AC$ (given)

AD is common.

$\angle ADB = \angle ADC = 90°$ ($AD \perp BC$)

$\therefore \triangle ABD \equiv \triangle ACD$ (RHS)

$\therefore \angle BAD = \angle CAD$ (matching angles of congruent triangles)

$\therefore AD$ bisects $\angle BAC$.

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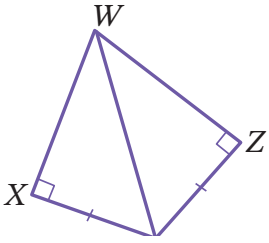
EXERCISE 13.03 ANSWERS ON P. 663

Congruent triangle proofs U F R C

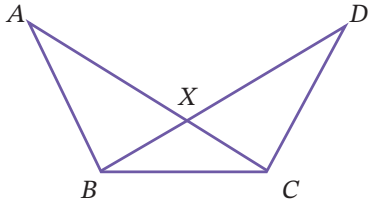
EXAMPLE 5

13.03

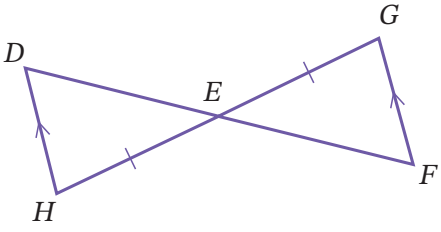
1 Prove that $\triangle WXY \equiv \triangle WZY$. R C



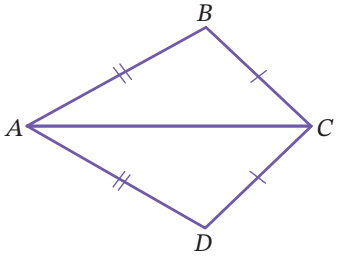
2 If $\angle ABC = \angle DCB$ and $AB = DC$, prove that $\triangle ABC \equiv \triangle DCB$. R C



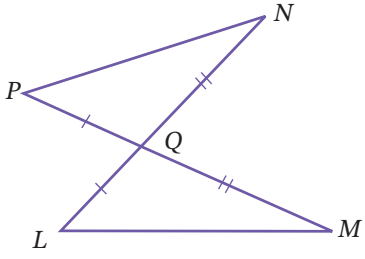
3 In the diagram, $EG = EH$ and $DH \parallel FG$. Show that $\triangle DEH \equiv \triangle FEG$. R C



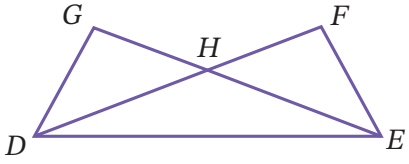
4 For this kite $ABCD$, prove that $\triangle ABC \equiv \triangle ADC$. R C



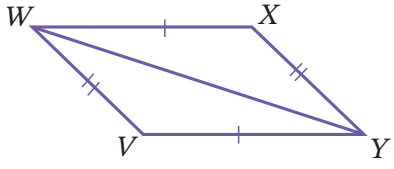
5 If $PQ = LQ$ and $NQ = MQ$, prove that $\triangle PQN \equiv \triangle LQM$. R C



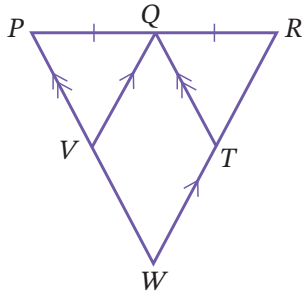
6 If $\angle DEG = \angle EDF$ and $GE = FD$, prove that $\triangle DEG \equiv \triangle EDF$. R C



7 Prove that $\triangle WXY \equiv \triangle YVW$. R C



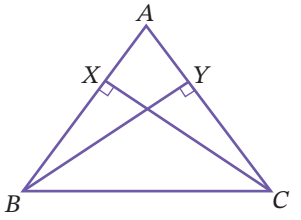
8 $PW \parallel QT$, $RW \parallel QV$ and $PQ = QR$. Prove that $\triangle PVQ \equiv \triangle QTR$. R C



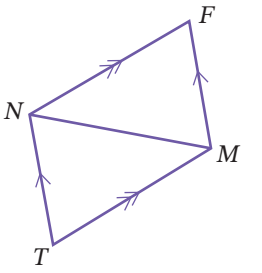
Foundation | Standard | Complex

PATH

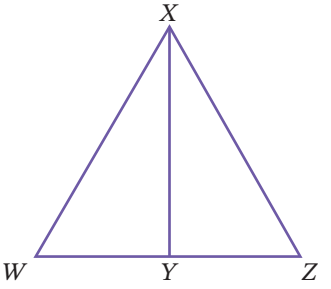
9 If $CX \perp AB$, $BY \perp AC$ and $XC = YB$, prove that $\triangle BCX \equiv \triangle CBY$. R C



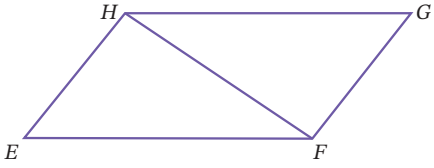
10 Prove that $\triangle FNM \equiv \triangle TMN$. R C



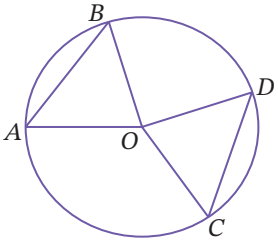
11 $XW = XZ$ in this isosceles triangle and Y is the midpoint of WZ. Prove that $\triangle WYX \equiv \triangle ZYX$. R C



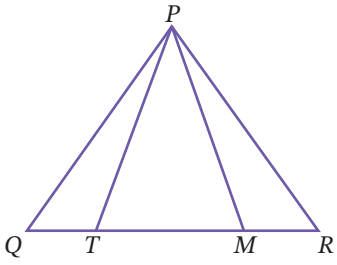
12 The opposite sides are equal for this quadrilateral. Prove that $\triangle EFH \equiv \triangle GHF$. R C



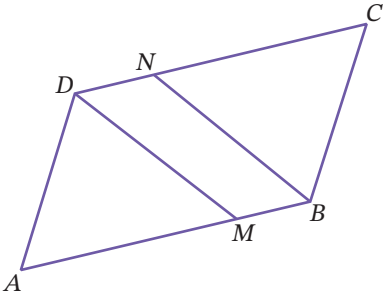
13 O is the centre of the circle and $AB = CD$. Prove that $\triangle AOB \equiv \triangle COD$. R C



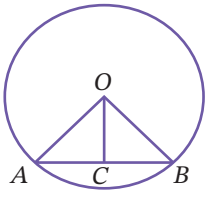
14 $\angle Q = \angle R$ in this isosceles triangle. If $QT = RM$, prove that $\triangle PQT \equiv \triangle PRM$. R C



15 $ABCD$ is a parallelogram and $DN = BM$. Prove that $\triangle AMD \equiv \triangle CNB$. R C



16 O is the centre of the circle and $OC \perp AB$. Prove that $\triangle OAC \equiv \triangle OBC$. R C



Foundation Standard Complex

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PATH

EXAMPLE 6

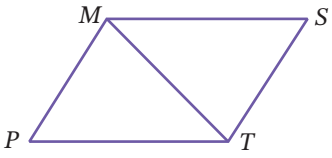
13.03

17 In the quadrilateral $MSTP$, opposite sides are equal. Prove that: R C

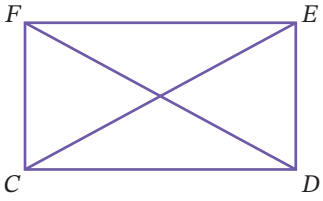
a $\triangle TPM \equiv \triangle MST$

b $\angle PMT = \angle STM$

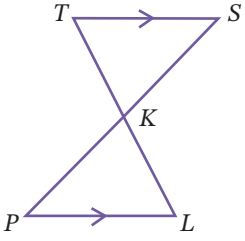
c $PM \parallel ST$



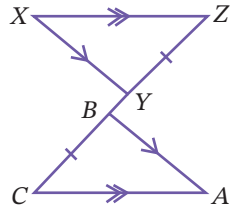
18 $CDEF$ is a rectangle. Prove that $\triangle FED \equiv \triangle EFC$ and that the diagonals of the rectangle are equal in length. R C



19 $TS \parallel PL$ and K is the midpoint of TL. Prove that $\triangle TSK \equiv \triangle LPK$. R C



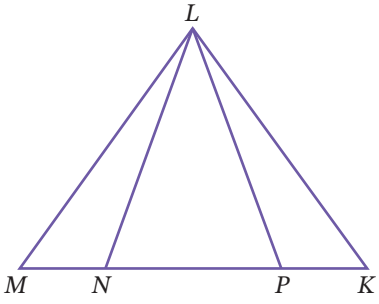
20 $AC \parallel XZ$, $XY \parallel AB$ and $BC = YZ$. R C



a Prove that $\angle XYZ = \angle ABC$.

b Hence prove that $\triangle XYZ \equiv \triangle ABC$.

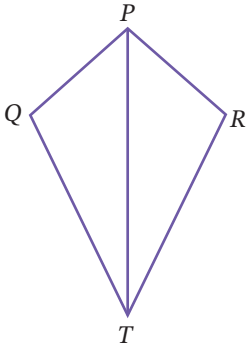
21 $LM = LK$ in this isosceles triangle LMK, and $MN = KP$. Prove that: R C



a $\triangle LMN \equiv \triangle LKP$

b $\triangle LNP$ is isosceles.

22 For this kite, prove that: R C



a $\triangle PQT \equiv \triangle PRT$

b $\angle Q = \angle R$

c PT bisects $\angle QTR$.

Foundation Standard Complex

13.04 Using congruence to prove geometrical properties

PATH

Properties of triangles and quadrilaterals can be proved using the congruence tests.

Video
Congruent triangles proofs

Worksheet
Proving properties of quadrilaterals

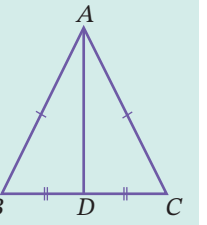
Example 7

$\triangle ABC$ is an isosceles triangle. AD bisects BC.

a Prove that $\triangle ABD \equiv \triangle ACD$.

b Hence prove that in an isosceles triangle, the angles opposite the equal sides are equal.

c Prove that $AD \perp BC$.



Solution

a In $\triangle ABD$ and $\triangle ACD$:

$AB = AC$ ($\triangle ABC$ is isosceles)

AD is common.

$BD = CD$ (AD bisects BC)

$\therefore \triangle ABD \equiv \triangle ACD$ (SSS)

b $\therefore \angle B = \angle C$ (matching angles of congruent triangles)

$\therefore$ The angles opposite the equal sides in an isosceles triangle are equal.

c $\angle ADB = \angle ADC$ (matching angles of congruent triangles)

$\therefore \angle ADB = \angle ADC = 90°$ (angles on a straight line)

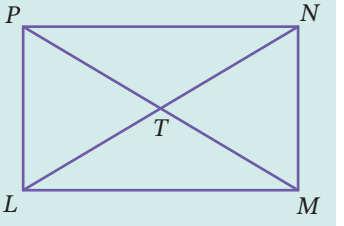
$\therefore AD \perp BC$.

Video
Proving properties of a rectangle

Example 8

a If $LMNP$ is a rectangle, prove that $\triangle PNT \equiv \triangle MLT$.

b Prove that the diagonals of a rectangle bisect each other.



Solution

a In $\triangle PNT$ and $\triangle MLT$:

$PN = ML$ (opposite sides of a rectangle)

$\angle PNT = \angle MLT$ (alternate angles on parallel lines, $PN \parallel ML$)

$\angle PTN = \angle MTL$ (vertically opposite angles)

$\therefore \triangle PNT = \triangle MLT$ (AAS)

b $\therefore PT = MT$ and $NT = LT$ (matching sides of congruent triangles)

$\therefore T$ is the midpoint of the diagonals LN and MP.

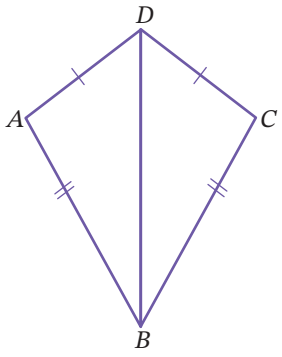
$\therefore$ The diagonals of a rectangle bisect each other.

EXERCISE 13.04 ANSWERS ON P. 664

Using congruence to prove geometrical properties U F R C

EXAMPLE 7

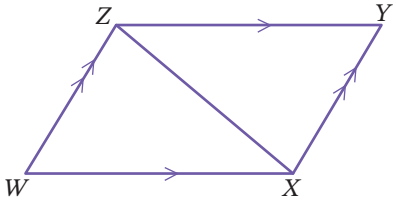
1 *ABCD* is a kite so adjacent sides are equal. R C



- **a** Prove that $\triangle ABD \equiv \triangle CBD$.
- **b** Prove that $\angle A = \angle C$.
- **c** Prove that diagonal *DB* bisects $\angle ADC$ and $\angle ABC$.
- **d** Copy the diagram and draw the other diagonal *AC*, intersecting *DB* at point *X*.
- **e** Prove that $\triangle DAX \equiv \triangle DCX$.
- **f** Prove that diagonal *DB* bisects diagonal *AC*.
- **g** Prove that $DB \perp AC$.

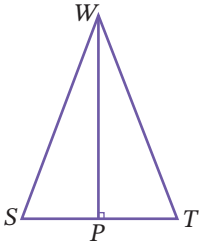
EXAMPLE 8

2 *WXYZ* is a parallelogram with opposite sides parallel. R C



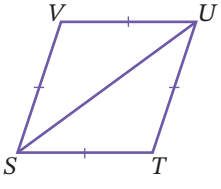
- **a** Copy the diagram into your book.
- **b** Prove that $\triangle WXZ \equiv \triangle YZX$.
- **c** Prove that $\angle W = \angle Y$.
- **d** Draw the other diagonal *WY* and prove that $\triangle WXY \equiv \triangle YZW$.
- **e** Prove that $\angle WXY = \angle YZW$.
- **f** What angle property of a parallelogram does this prove?

3 In the diagram, $\angle S = \angle T$ and $WP \perp ST$. R C



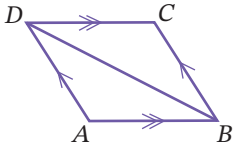
- **a** Prove that $\triangle SPW \equiv \triangle TPW$.
- **b** Prove that the sides opposite equal angles in a triangle are equal.

4 *STUV* is a rhombus, so all sides are equal. R C



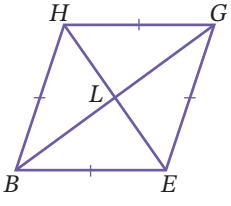
- **a** Prove that $\triangle VUS \equiv \triangle TUS$.
- **b** Prove that the diagonal *US* bisects the angles $\angle VUT$ and $\angle VST$.

5 *ABCD* is a parallelogram with opposite sides parallel. R C



- **a** Prove that $\triangle ABD \equiv \triangle CDB$.
- **b** Prove that opposite sides of a parallelogram are equal.

6 *BEGH* is a rhombus (a parallelogram with equal sides) whose diagonals *BG* and *EH* intersect at *L*. R C



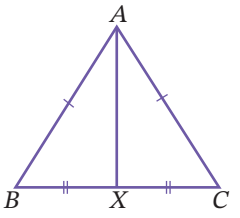
- **a** Prove that $\triangle BEL \equiv \triangle GHL$.
- **b** Prove that the diagonals of a rhombus bisect each other.
- **c** $\triangle BEH$ is isosceles, so which 2 angles are equal?
- **d** Hence prove that $\triangle BEL \equiv \triangle BHL$.
- **e** Hence prove that the diagonals of a rhombus cross at right angles.

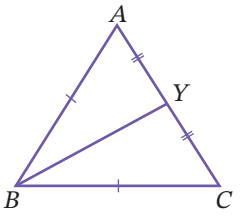
13.04

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PATH

7 $\triangle ABC$ is an equilateral triangle ($AB = BC = AC$). X is the midpoint of BC. R C



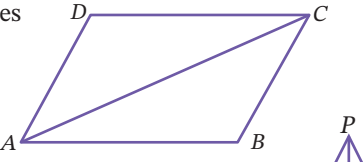


a Prove that $\triangle ABX \equiv \triangle ACX$ and $\angle B = \angle C$.

b In the second diagram, $\triangle ABC$ is redrawn so that Y is the midpoint of AC.
Prove that $\triangle BAY \equiv \triangle BCY$ and $\angle A = \angle C$.

c Calculate the sizes of the 3 angles of $\triangle ABC$.

8 $ABCD$ is a quadrilateral in which opposite sides are equal. R C

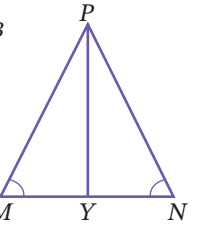


a Prove that $\triangle ACD \equiv \triangle ACB$.

b Prove that $AB \parallel CD$ and $AD \parallel BC$.

c What type of quadrilateral is $ABCD$?

9 In $\triangle PMN$, $\angle M = \angle N$ and YP bisects $\angle MPN$. R C

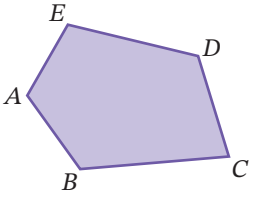


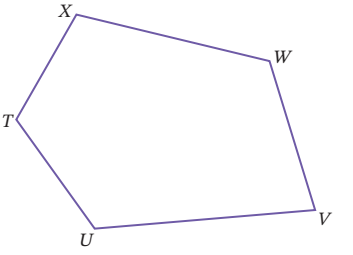
a Prove that $\triangle PMY \equiv \triangle PNY$.

b Prove that $MY = NY$ and $PY \perp MN$.

Investigation

Same shape, different size





1 **a** Consider the pentagons $ABCDE$ and $TUVWX$. Do they look similar?

b What do you think is meant by the word 'similar'?

c Copy and complete this table by measuring each angle and side in the 2 'similar' pentagons.

d What do you notice about the sizes of the matching angles?

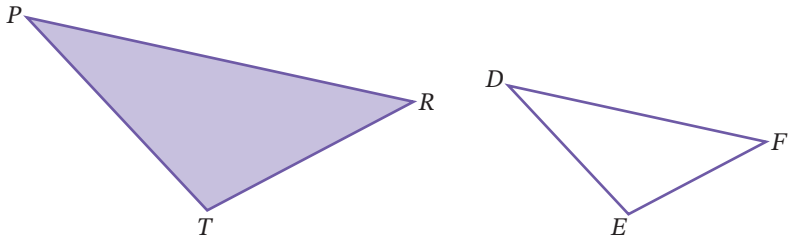
e What do you notice about the sizes of the matching sides?

Angles		Sides (in mm)	
$A =$	$T =$	$AB =$	$TU =$
$B =$	$U =$	$BC =$	$UV =$
$C =$	$V =$	$CD =$	$VW =$
$D =$	$W =$	$DE =$	$WX =$
$E =$	$X =$	$AE =$	$TX =$

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2 $\triangle PTR$ has been reduced to produce the similar triangle $\triangle DEF$.



a List the 3 pairs of matching angles and measure the size of each one.

b List the 3 pairs of matching sides and measure the length of each one.

3 Copy and complete:

a For similar figures, m_____ angles are e_____.

b Similar figures have the s_____ shape, but not the same s_____.

Similar figures 13.05

Similar figures have the same shape but are not necessarily the same size.

When a photograph is enlarged or reduced, all parts of the picture remain the same shape, but all lengths and distances will be respectively larger or smaller than the corresponding lengths and distances on the original.



iStock.com/aluxum

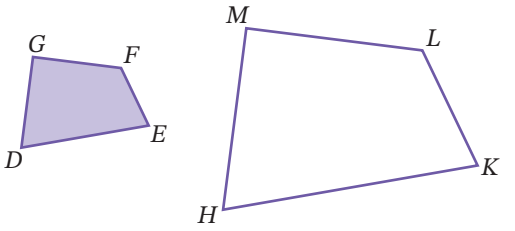
When a figure is enlarged or reduced, a **similar figure** is created. The original figure is called the **original**, while the enlarged or reduced figure is called the **image**.

Worksheets
A page of similar figures
A page of congruent and similar figures
Enlargements and reductions
Enlarging a logo

Puzzle
Cartoon enlargement

The scale factor

The **scale factor** describes by how much a figure has been enlarged or reduced. For example, if the quadrilateral *DEFG* is enlarged by doubling each side, the scale factor is 2. Quadrilaterals *DEFG* and *HKLM* are similar. $\angle D$ matches $\angle H$, $\angle E$ matches $\angle K$, and so on.



Original Image

$HK = 2 \times DE$

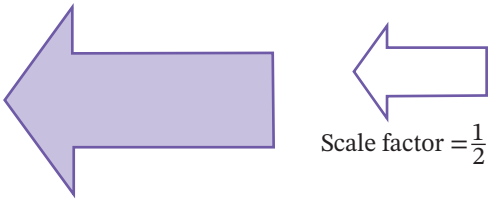
$KL = 2 \times EF$

$LM = 2 \times FG$

$HM = 2 \times DG$

$\therefore$ Scale factor $= 2$

In a similar manner, if the arrow figure is reduced by halving each side, the scale factor is $\frac{1}{2}$.



Original Image

Scale factor

$$\text{Scale factor} = \frac{\text{image length}}{\text{original length}}$$

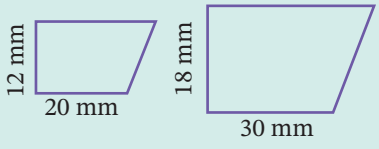
- If the scale factor is greater than 1, then the image is an enlargement.
- If the scale factor is between 0 and 1, then the image is a reduction.

Example 9

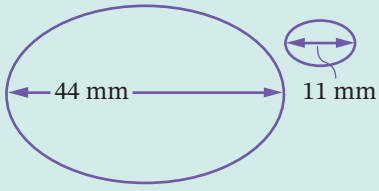
Find the scale factor for each pair of similar figures.

In all questions, assume the left figure is the original and the right figure the image.

a



b



Solution

a Scale factor $= \frac{30}{20}$ (or $\frac{18}{12}$)

$= 1\frac{1}{2}$

$\frac{\text{image length}}{\text{original length}}$

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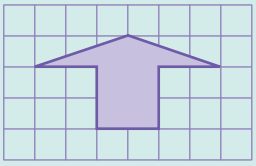
b Scale factor $= \frac{11}{44}$ $\left(\frac{\text{image length}}{\text{original length}}\right)$

$= \frac{1}{4}$

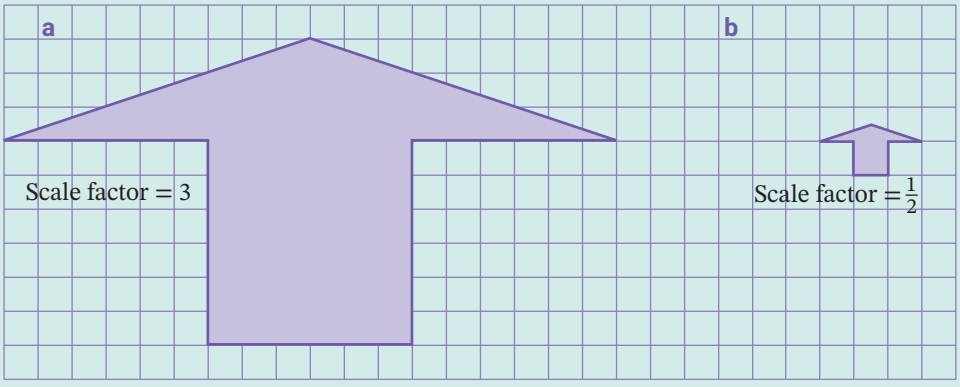
Example 10

Construct the image of the figure shown if the scale factor is:

a 3 **b** $\frac{1}{2}$



Solution



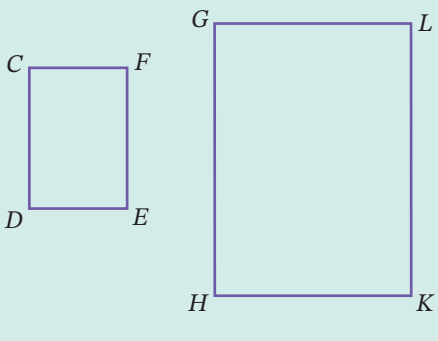
The similarity symbol ~

The symbol for 'is similar to' is '~' or '|||'. As with congruence notation, we must make sure that the vertices (or angles) of the similar figures are written in matching order.

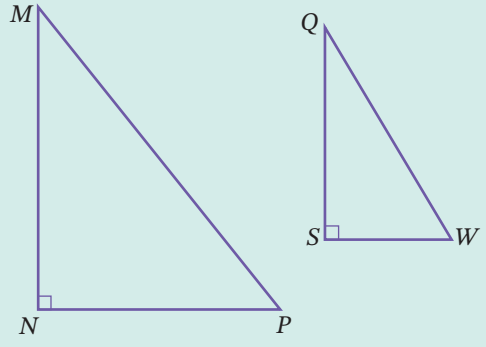
Example 11

Test whether each pair of figures are similar by measuring them.

a



b



Solution

a The rectangle $GHKL$ is an enlargement of rectangle $CDEF$.

$$\text{Scale factor} = \frac{GH}{CD} = \frac{36}{18} = 2$$

$$\frac{HK}{DE} = \frac{26}{13} = 2$$

$$\therefore \frac{GH}{CD} = \frac{HK}{DE}$$

$\therefore CDEF \,|||\, GHKL$ Matching order of vertices

b $\triangle QWS$ is not a reduction of $\triangle MPN$.

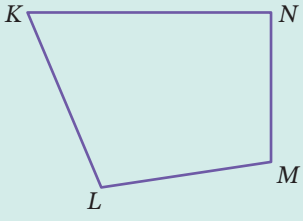
$$\frac{QS}{MN} = \frac{28}{40} = \frac{7}{10} \qquad \frac{SW}{NP} = \frac{16}{32} = \frac{1}{2}$$

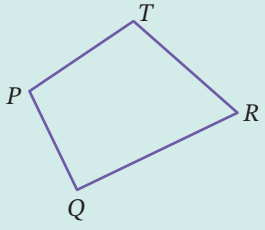
$$\frac{QS}{MN} \neq \frac{SW}{NP}$$

$\therefore \triangle MNP$ is not similar to $\triangle QSW$.

Example 12

The 2 quadrilaterals $KLMN$ and $PQRT$ are similar.





a List all pairs of matching sides and matching angles.

b Use the correct notation to write a similarity statement relating these 2 quadrilaterals.

Solution

a By rotating $KLMN$ and superimposing it over $PQRT$, its shape can be matched with $PQRT$.

The pairs of matching sides are:	The pairs of matching angles are:
KN and QR	$\angle K$ and $\angle R$
MN and PQ	$\angle N$ and $\angle Q$
ML and PT	$\angle M$ and $\angle P$
LK and TR.	$\angle L$ and $\angle T$.

b $\angle K$ matches with $\angle R$, $\angle L$ matches with $\angle T$, $\angle M$ matches with $\angle P$, $\angle N$ matches with $\angle Q$.

$\therefore KLMN \sim RTPQ$ Matching order of vertices

EXERCISE 13.05 ANSWERS ON P. 664

Similar figures UFRC

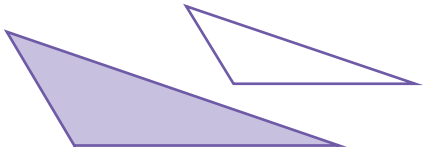
1 By measurement, find the scale factor for each pair of similar figures.

EXAMPLE 9

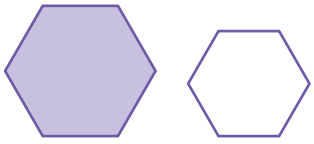
a



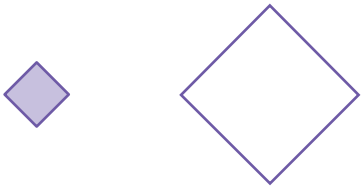
b



c



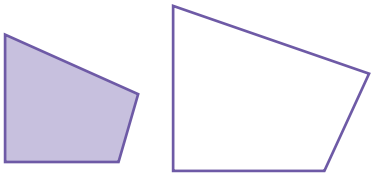
d



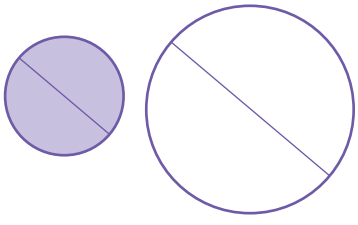
e



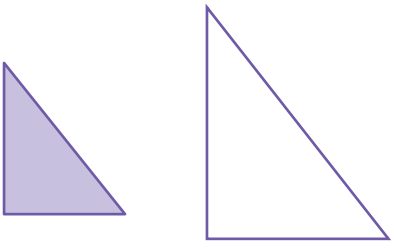
f



g



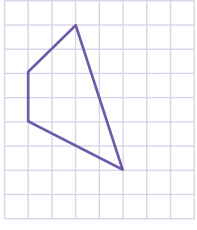
h



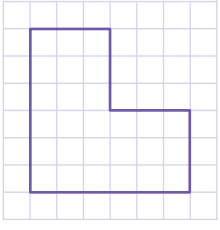
2 Copy each figure onto grid paper and draw its image using the given scale factor.

EXAMPLE 10

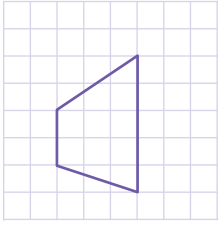
a Scale factor = 2



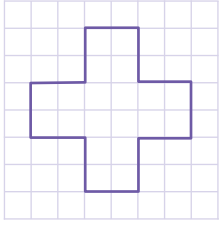
b Scale factor = $\frac{1}{3}$



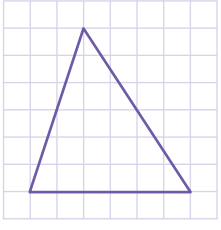
c Scale factor = 2



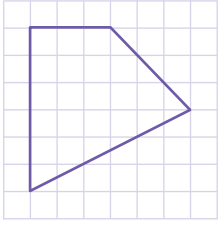
d Scale factor = 3



e Scale factor = $\frac{1}{2}$



f Scale factor = $\frac{2}{3}$



Worksheet
5 mm grid paper

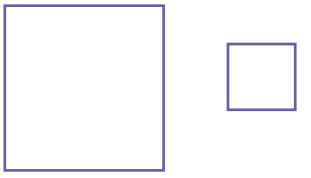
13.05

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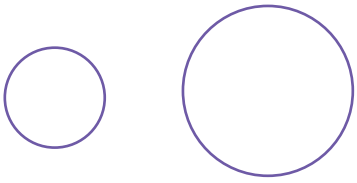
EXAMPLE 11

3 Are each pair of figures similar?

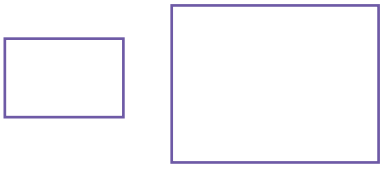
a



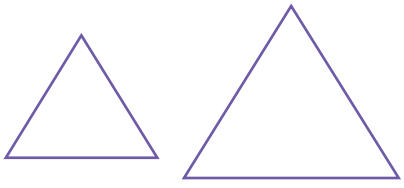
b



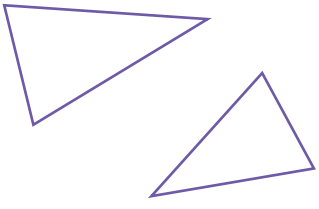
c



d



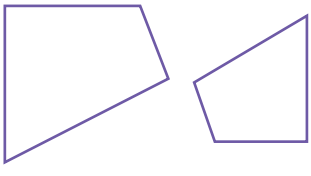
e



f



g



h



EXAMPLE 12

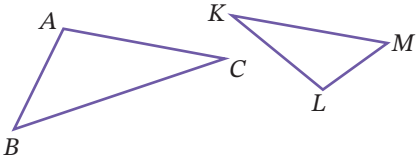
4 For each pair of similar figures: R C

i list all pairs of matching angles

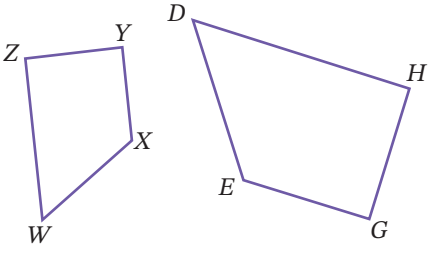
ii list all pairs of matching sides

iii use the correct notation ('~') to write a similarity statement relating them.

a

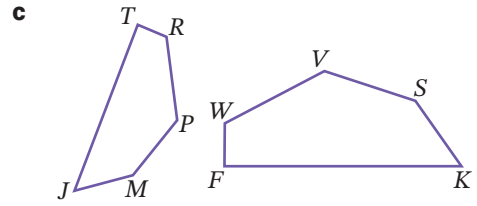


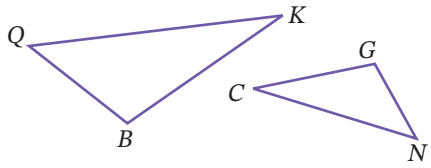
b

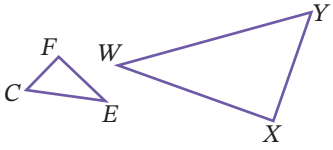


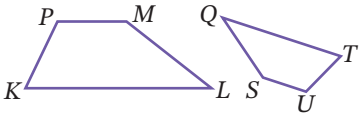
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13.05

Quiz
Mental skills 13

Mental skills 13: Maths without calculators ANSWERS ON P. 665

Simplifying fractions and ratios

When simplifying a fraction or a ratio, look for a common factor to divide into both the numerator and the denominator, preferably the highest common factor (HCF). Remember the divisibility tests:

A number is divisible by ...	if ...
2	it is even (its last digit is 2, 4, 6, 8 or 0)
3	the sum of its digits is divisible by 3
4	its last 2 digits form a number divisible by 4
5	its last digit is 0 or 5
6	it is even and the sum of its digits is divisible by 3
9	the sum of its digits is divisible by 9
10	its last digit is 0

1 Study each example.

a Simplify $\frac{27}{45}$.

$\frac{27^{9}}{45_{15}} = \frac{9}{15}$ dividing numerator and denominator by 3

$\frac{9^{3}}{15_{5}} = \frac{3}{5}$ dividing numerator and denominator by 3 again

Note: This fraction could be simplified in one step if you divided by 9, the highest common factor (HCF) of 27 and 45.

b Simplify $\frac{160}{400}$.

$\frac{160}{400} = \frac{16}{40}$ dividing numerator and denominator by 10

$\frac{16^{2}}{40_{5}} = \frac{2}{5}$ dividing numerator and denominator by 8

Note: This fraction could be simplified in one step if you divided by 80, the HCF of 160 and 400.

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c Simplify 24 : 36.

$24:36 = \cancel{24}^{4} : \cancel{36}^{6} = 4:6$ dividing both terms by 6

$\cancel{4}^{2} : \cancel{6}^{3} = 2:3$ dividing both terms by 2

Note: This fraction could be simplified in one step if you divided by 12, the HCF of 160 and 400.

d Simplify 135 : 90.

$135:90 = \cancel{135}^{27} : \cancel{90}^{18} = 27:18$ dividing both terms by 5

$\cancel{27}^{3} : \cancel{18}^{2} = 3:2$ dividing both terms by 9

e Calculate $\frac{3}{8} \times \frac{2}{15}$ in simplest form.

$\frac{3}{8} \times \frac{2}{15} = \frac{3}{\cancel{8}_{4}} \times \frac{\cancel{2}^{1}}{15}$ dividing 2 and 8 by 2

$= \frac{\cancel{3}^{1}}{8} \times \frac{2}{\cancel{15}_{5}}$ dividing 3 and 15 by 3

$= \frac{1}{20}$

f What fraction is 36 minutes of 1 hour?

$\frac{36}{1\text{ h}} = \frac{36\text{ min}}{60\text{ min}} = \frac{3}{5}$

2 Now simplify each fraction or ratio.

a $\frac{10}{15}$	**b** $\frac{16}{20}$	**c** $\frac{30}{42}$	**d** $\frac{8}{16}$
e $\frac{20}{80}$	**f** $\frac{6}{36}$	**g** $\frac{20}{24}$	**h** $\frac{12}{30}$
i 20 : 36	**j** 25 : 45	**k** 18 : 40	**l** 28 : 35
m 27 : 21	**n** 16 : 12	**o** $\frac{5}{6} \times \frac{18}{25}$	**p** $\frac{12}{50} \times \frac{10}{21}$

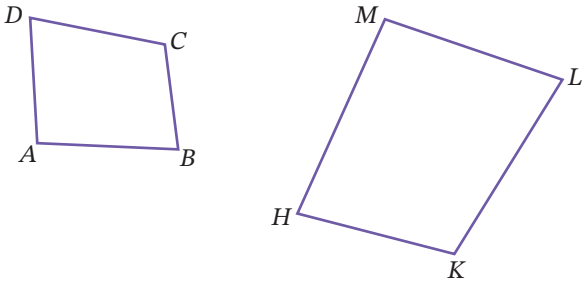
3 Express each as a simplified fraction.

a 425 g of 1 kg	**b** 8 months of 1 year	**c** 64 cm of 1 m
d 750 mL of 3 L	**e** 10 hours of 2 days	**f** 80c of $10

Investigation

Properties of similar figures

1 The quadrilaterals *ABCD* and *HKLM* are similar.



a Measure the sizes of each pair of matching angles. What do you notice?

b Measure the lengths of each pair of matching sides.

c Calculate each ratio of matching sides, as a decimal correct to one decimal place.

i $\frac{MH}{AB}$ **ii** $\frac{HK}{BC}$ **iii** $\frac{KL}{CD}$ **iv** $\frac{ML}{AD}$

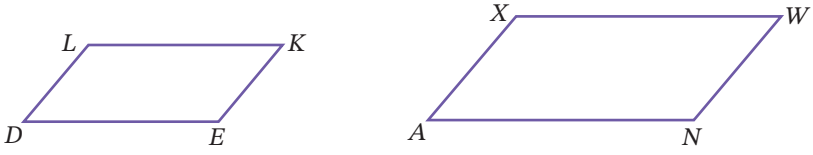
What do you notice?

d Copy and complete.

i The matching a_____ of similar figures are e_____.

ii Matching s_____ of similar figures are in the same r_____.

2 Consider the parallelograms *DEKL* and *ANWX*.



a Measure the sizes of each pair of matching angles. Are they equal?

b Measure the lengths of each pair of matching sides.

c Calculate each ratio of matching sides, as a decimal correct to one decimal place:

i $\frac{AN}{DE}$ **ii** $\frac{AX}{DL}$

Are they equal?

d *DEKL* is not similar to *ANWX*. Explain why.

13.06 Finding sides in similar figures

Skillsheet
Finding sides in similar triangles

Worksheets
Finding sides in similar figures

Congruence and similarity review

Puzzle
Similar triangles

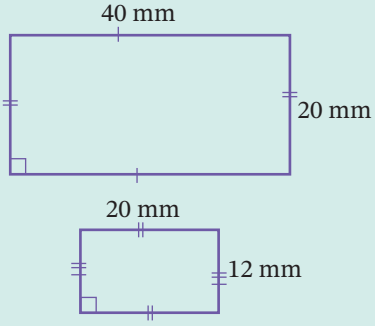
Properties of similar figures

- **Matching angles** are equal
- **Matching sides** are in the same ratio

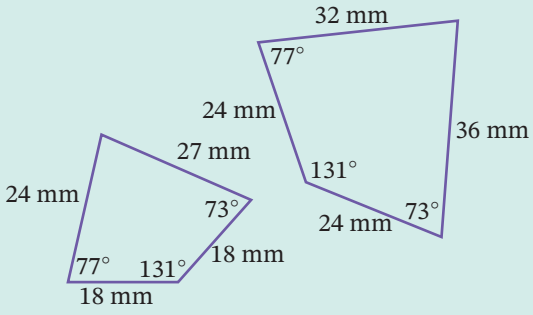
Example 13

Test whether each pair of figures are similar.

a



b



Solution

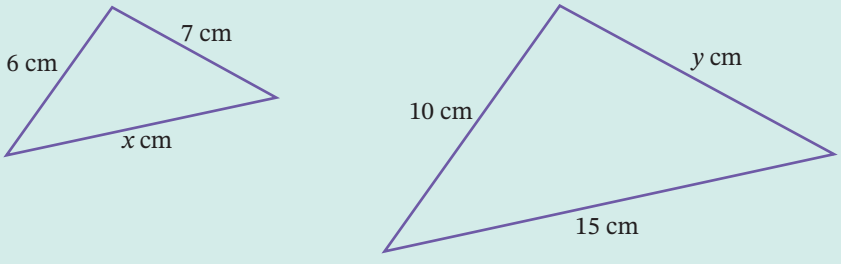
a For the 2 rectangles, matching angles are equal (90°), but the ratios of the matching sides are not equal $\left(\frac{20}{40}=\frac{1}{2} \text{ but } \frac{12}{20}=\frac{3}{5}\right)$.
∴ The rectangles are not similar.

b For the 2 quadrilaterals, matching angles are equal and the ratios of matching sides are equal $\left(\frac{32}{24}=\frac{36}{27}=\frac{24}{18}=\frac{4}{3}\right)$.
∴ The quadrilaterals are similar.

Video
Finding an unknown side in similar figures

Example 14

These 2 triangles are similar. Find the value of x and y.



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Solution

Since the triangles are similar, the ratios of matching sides are equal.

$\frac{x}{15} = \frac{6}{10}$

$x = \frac{6}{10} \times 15$

$= 9$

$\frac{y}{7} = \frac{10}{6}$

$y = \frac{10}{6} \times 7$

$= 11\frac{2}{3}$

Alternative method:

Scale factor $= \frac{10}{6} = \frac{5}{3}$

$x = 15 \div \frac{5}{3}$

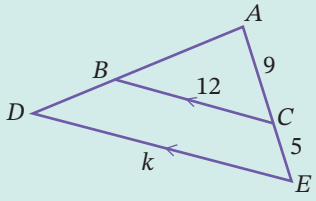
$= 9$

$y = 7 \times \frac{5}{3}$

$= 11\frac{2}{3}$

Example 15

$\triangle ABC \sim \triangle ADE$. Find the value of k.



Solution

$\frac{DE}{BC} = \frac{AE}{AC}$ Ratios of matching sides are equal

$\frac{k}{12} = \frac{14}{9}$ $AE = 9 + 5 = 14$

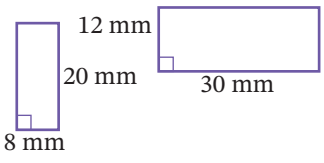
$k = \frac{14}{9} \times 12 = 18\frac{2}{3}$

EXERCISE 13.06 ANSWERS ON P. 665

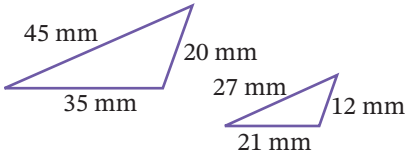
Finding sides in similar figures U F PS R

1 Test whether each pair of figures are similar. R

a



b



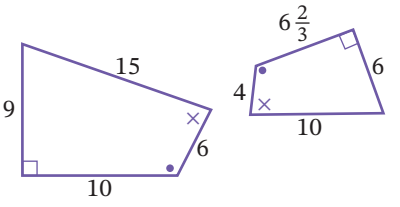
13.06

Puzzle
Nested similar triangles

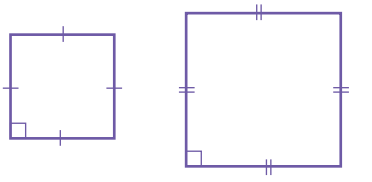
EXAMPLE 13

Foundation Standard Complex

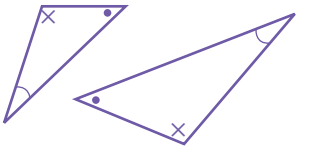
c



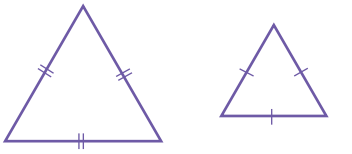
d



e



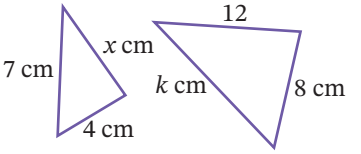
f



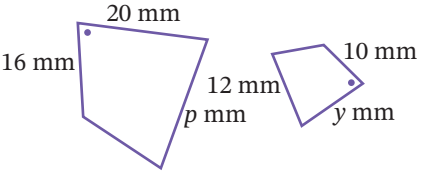
EXAMPLE 14

2 Find the value of each variable in each pair of similar figures. R

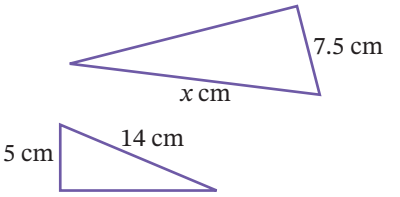
a



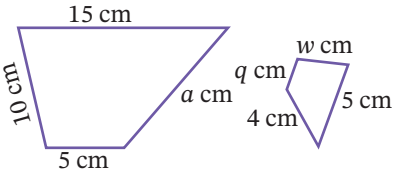
b



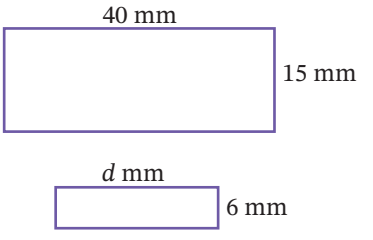
c



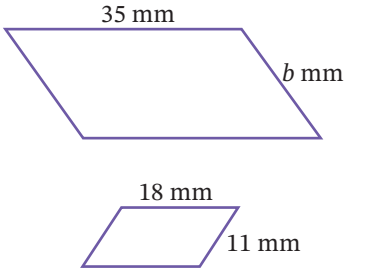
d



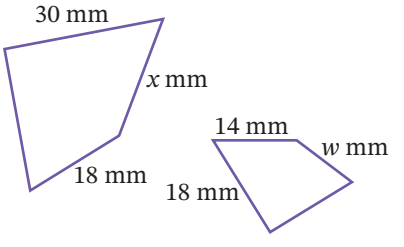
e



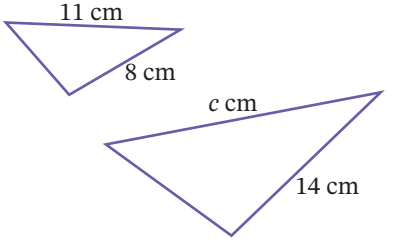
f



g



h



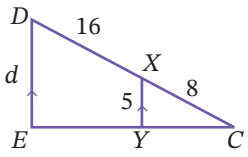
Foundation Standard Complex

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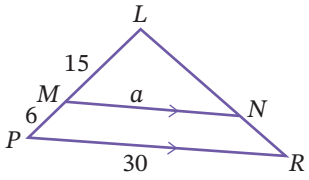
EXAMPLE 15

13.06

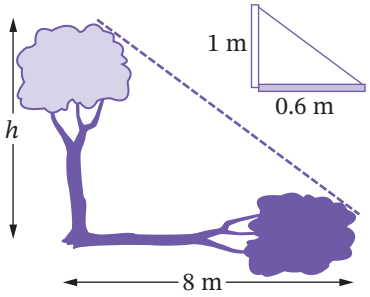
3 $\triangle CED \sim \triangle CYX$. Find the value of d. R



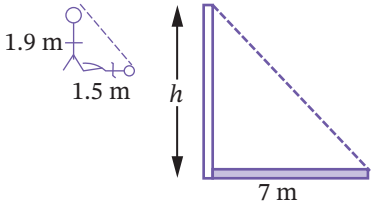
4 $\triangle LMN \sim \triangle LPR$. Find the value of a. R



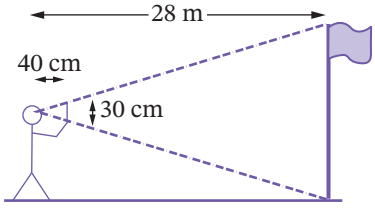
5 A metre stick casts a shadow of length 0.6 m when a tree casts a shadow of 8 m. Find the height h of the tree. R



6 Ayon, who is 1.9 m tall, casts a 1.5 m long shadow. At the same time, a light pole casts a shadow 7 m long. What is the height of the light pole, correct to one decimal place? R

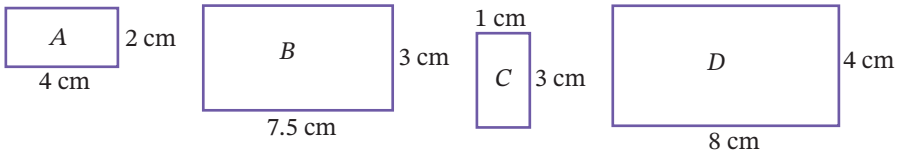


7 Emily holds up a small 40 cm long stick from her eye so that it appears to be the same height as the flagpole. She is standing 28 m from the flagpole and the stick is 30 cm long. Find the height of the flagpole. PS R



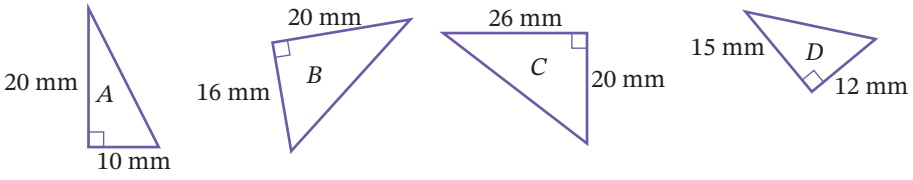
8 For each set of figures, find the pair of similar figures and the scale factor between them. R

a

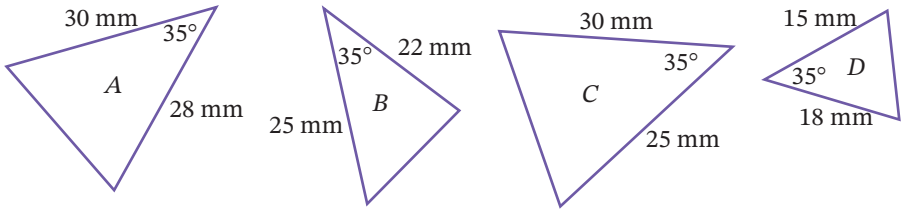


Foundation Standard Complex

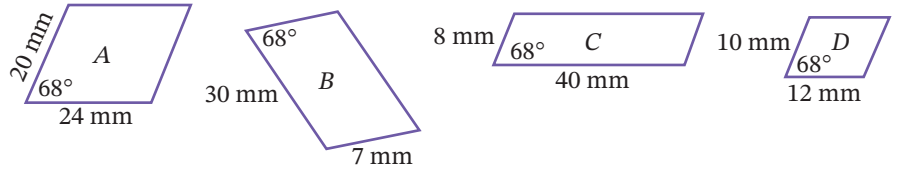
b



c



d



Investigation

Are all shapes similar?

Work in groups of 3 or 4 to complete this activity.

You will need: a ruler, a pencil, scissors.

1 **a** Draw 2 equilateral triangles, using a different side length for each triangle.

b Are the 2 equilateral triangles similar? Give reasons.

c Compare your triangles with those of other members of your group. Are all equilateral triangles similar? Give reasons.

2 **a** Draw a pair of similar isosceles triangles.

b Are all isosceles triangles similar? Give reasons.

3 **a** Draw a pair of rectangles that are similar.

b Draw a pair of rectangles that are *not* similar.

c When are rectangles similar? Discuss your answer with other students.

4 State whether the following statements are true or false. Give reasons for your answers. (Making a few sketches will help.)

a All parallelograms are always similar.

b All circles are always similar.

c All squares are always similar.

d All rhombuses are always similar.

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Scale drawings

13.07

A **scale drawing** accurately represents a larger or smaller object such that the lengths and distances on it are in the same ratio as the real lengths and distances. This means that the scale drawing is either an enlargement or reduction of the real object. The scale factor used is called the **scale** of the diagram.

A scale may be represented in different ways.

- 1 cm = 10 m or 1 cm to 10 m (a pair of corresponding measurements)
- 1 : 50 (a ratio)
- 

Scaled length

The scale on a scale drawing is written as a ratio of **scaled length : real length**, where **scaled length** is the length on the diagram.

For example, a scale of 1 : 100 means that the real lengths are 100 times larger than the lengths on the diagram.

Example 16

A scale drawing of a rectangular block of land has been drawn. What scale has been used?





Shutterstock.com/jax10289

Solution

By measurement, the length and width of the scale drawing are 35 mm and 20 mm.

Scale = 35 mm : 35 m (or 20 mm : 20 m) scaled length : real length

= 35 mm : 35 000 mm (or 20 mm : 20 000 m) 1 m = 1000 mm

= 1 : 1000

The actual length and width are 1000 times the scaled length and width.

Worksheet
Problems involving scale drawings

Technology
Converting map scales to ratios

Spreadsheet
Converting map scales to ratios

Videos
Scale drawings

Jai Singh (sundial)

Modelling the Spitfire

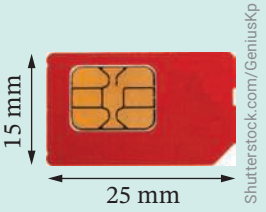
Queen Hatshepsut's ship

The history of the golden ratio

Fractals: The Koch snowflake

Example 17

A SIM card for a mobile phone has a length of 25 mm and a width of 15 mm. A scale diagram of the SIM card is shown below.



Shutterstock.com/GeniusKp



By measurement, determine what scale has been used.

Solution

By measurement, the length and width of the scale diagram are 75 mm and 45 mm respectively.

Scale = 75 mm : 25 mm (or 45 mm : 15 mm) scaled length : real length

= 3 : 1

The actual length and width are $\frac{1}{3}$ of the scaled length and width.

Example 18

The scale on a map is 1 : 500 000. If the distance from Campbelltown to Sydney on the scale drawing is 9 cm, calculate the actual distance from Campbelltown to Sydney.

Solution

Scaled distance = 9 cm

Actual distance = 9 × 500 000 cm — 500 000 times larger

= 4 500 000 cm

= 45 000 m — 1 m = 100 cm

= 45 km — 1 km = 1000 m

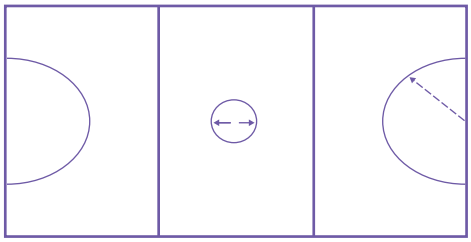
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EXERCISE 13.07 ANSWERS ON P. 665

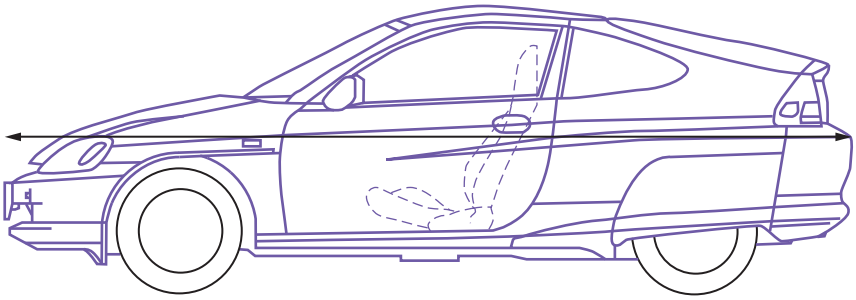
Scale drawings **U F R**

EXAMPLE 16

1 If the length of a netball court is 30.5 m, what scale has been used on this diagram?



2 What scale has been used on this diagram of a car if its actual length is 3.92 m?



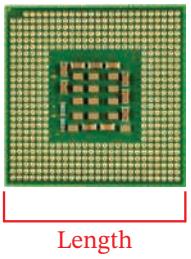
EXAMPLE 17

3 The picture of the Giant Brown Bull Ant has been enlarged. If its actual length is 25 mm, what scale has been used?



Nature Picture Library/Kim Taylor

4 A scale image of a microchip is shown. Determine the scale that has been used if its real length is 8 mm.



iStock.com/Fotokot197

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5 The Q1 Tower on the Gold Coast, Queensland, is 322 m high and is Australia's tallest building. What scale has been used in this photo if the scaled height is 40 mm?



Shutterstock.com/Steven Bostock

6 This photo of the Sydney Harbour Bridge has a scale of 1 : 9000. By measurement and calculation, find: R

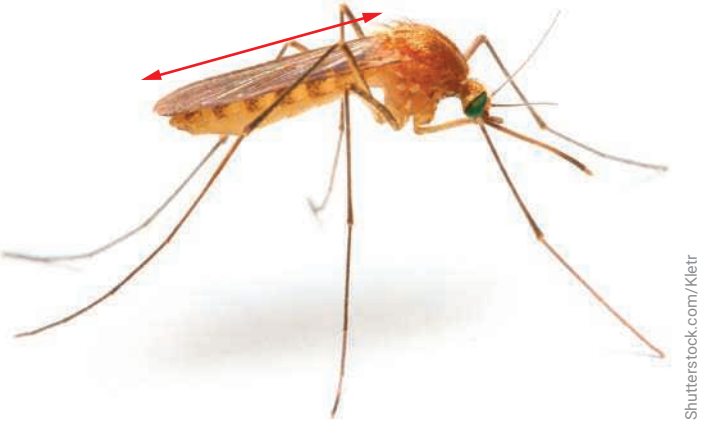
a the height of the top of the bridge above the water



Shutterstock.com/ErikSandstrom

b the length of the arch span of the bridge.

7 This is a photograph of a mosquito, at a scale of 15 : 1. Calculate, to the nearest 0.1 mm, the length of the wing. R



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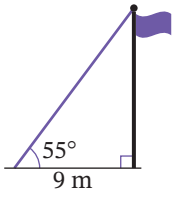
8 This box of cereal is 16 cm long and 22 cm tall. Construct a scale drawing of the front of the box, using a scale of 1 : 5.



Shutterstock.com/Nor Gal

9 **a** Construct a scale drawing of this flagpole using a scale of 1 : 300.

b Use the scale to calculate the actual height of the flagpole. R



13.07

EXAMPLE 18

10 The scale on a map of Australia is 1 : 10 000 000. If the distance from Perth to Adelaide on the map is 21.2 cm, calculate the actual distance in kilometres. PS R

11 Use the scale on this map to find, to the nearest kilometre, the distance from: R

a Nowra to Batemans Bay

b Bega to Moruya

c Mollymook/Ulladulla to Mogo

d Nowra to Eden.



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PATH

Technology
Area of similar shapes

Spreadsheet
Area of similar shapes

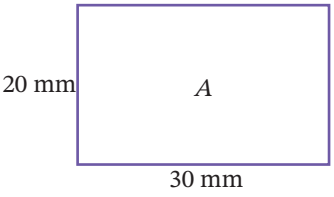
Investigation

Areas of similar figures

1 **a** The 2 rectangles A and B are similar. Explain why.

b What is the ratio of their matching sides (A to B)?

c What is the ratio of their areas (A to B)?



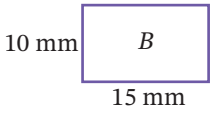
2 Triangles K and L are similar.

a What is the ratio of their matching sides (K to L)?

b Find the areas of both triangles.

c What is the ratio of their areas (K to L)?

d Copy and complete: When you double the lengths of a figure, the area is increased _____ times.



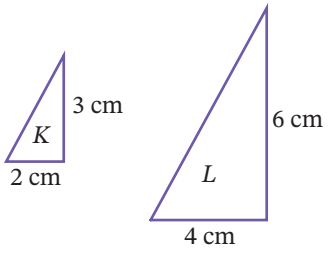
3 **a** What is the area of a rectangle with length 5 cm and width 4 cm?

b The length and width of the rectangle are both multiplied by 3. What is the area of the enlarged rectangle?

c What is the ratio of the matching sides of the 2 rectangles from parts **a** and **b**?

d What is the ratio of the areas of the 2 rectangles?

e Copy and complete: When you triple the lengths of a figure, the area is increased _____ times.



4 **a** What is the area of a circle with radius 2 cm?

b The radius of this circle is multiplied by 4.

c What is the area of the enlarged circle?

d What is the ratio of the areas of the 2 circles?

e Copy and complete: When you multiply the lengths of a figure by 4, the area is increased _____ times.

5 If the sides of a square are multiplied by 5, by how many times is the area increased?

6 If the sides of a figure are multiplied by k, by how many times is the area increased?

Technology

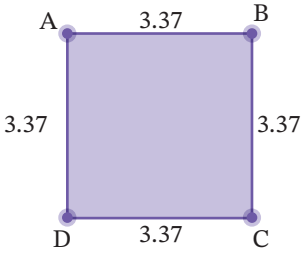
Areas of similar figures

For this activity, use dynamic geometry software to draw similar figures and compare their areas.

1 Construct a square. An example is shown.

2 Calculate the area of the square you have constructed in question 1.

3 Dilate (enlarge) the square by a scale factor of 2.



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4 Use the tools to:

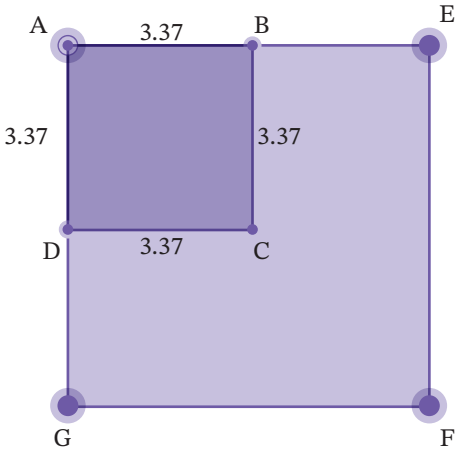
a measure the length of each side

b calculate the area of the enlarged square.

c What is the ratio of the sides $DC : GF$?

d What is the ratio of the areas of the squares $ABCD : AEFG$?

e Compare the 2 ratios, **c** and **d**. How are they related?



5 **a** Draw a rectangle with length 3 cm and width 1.5 cm. What is the area of the rectangle?

b Dilate the rectangle by a scale factor of 2 to enlarge the rectangle.

c Find the ratios of

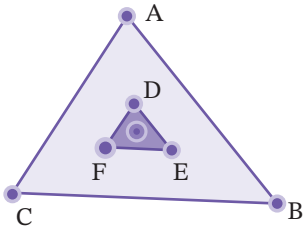
i sides $DC : GF$ **ii** sides $BC : EF$

iii areas of rectangles $ABCD : AEFG$.

d Compare the ratio of the sides to the ratio of the areas. How are they related?

6 **a** Draw a triangle with $AB = 8$ cm. It should look like triangle ABC shown.

b Dilate (reduce) the triangle by a scale factor of $\frac{1}{4}$ to reduce the triangle. Dilation can also be done by creating a point inside triangle ABC (it doesn't have to be one of the vertices, as shown in the diagram above).



c Find the ratios of:

i sides $AB : DE$ **ii** areas of triangles $ABC : DEF$

d Compare the ratio of the sides to the ratio of the areas. How are they related?

7 **a** Draw any polygon.

b Select **dilate** and either enlarge or reduce the polygon.

c Find the ratios of:

i matching sides of the similar shapes **ii** the areas of the similar shapes

d Compare the ratio of the sides to the ratio of the areas. How are they related?

13.08 Areas of similar figures

PATH

Areas of similar figures

If the matching sides of 2 similar figures are in the ratio $1 : k$, then their areas are in the ratio $1 : k^2$.

If the matching sides are in the ratio $m : n$, then their areas are in the ratio $m^2 : n^2$.

Example 19

a 2 similar triangles have sides in the ratio 1 : 5. Find the ratio of their areas.

b 2 similar parallelograms have sides in the ratio 3 : 1. Find the ratio of their areas.

Solution

a Ratio of sides $= 1 : 5$

Ratio of areas $= 1 : 5^2$ $\quad 1 : k^2$

$= 1 : 25$

b Ratio of sides $= 3 : 1$

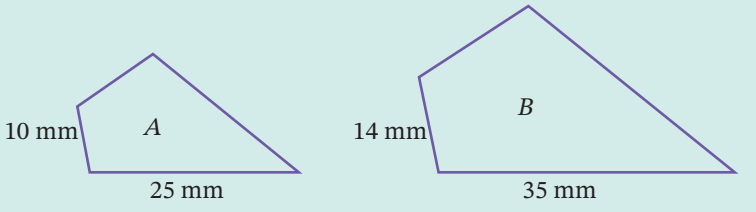
Ratio of areas $= 3^2 : 1$

$= 9 : 1$

Example 20

Video
Areas of similar figures

What is the ratio of the areas of the similar quadrilaterals shown?



Solution

Ratio of matching sides $(A \text{ to } B) = 25 : 35$ or $10 : 14$

$= 5 : 7$

Ratio of areas $(A \text{ to } B) = 5^2 : 7^2$

$= 25 : 49$

Example 21

Two similar trapeziums have their areas in the ratio 25 : 16.
Find the ratio of the lengths of the matching sides.

Solution

Ratios of areas $= m^2 : n^2 = 25 : 16$

$\therefore$ Ratio of sides $= m : n = \sqrt{25} : \sqrt{16} = 5 : 4$

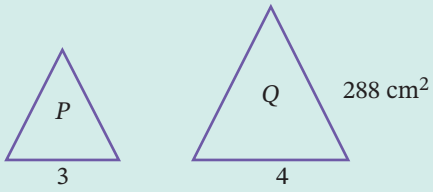
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Example 22

Two similar triangles P and Q have matching sides in the ratio 3 : 4.
If the area of the larger triangle is 288 cm^2, find the area of the smaller triangle.

Solution

Drawing a diagram of the problem is useful.



Ratio of matching sides = 3 : 4

Ratio of areas = $3^2 : 4^2 = 9 : 16$

$$\therefore \frac{\text{Area of } P}{288} = \frac{9}{16}$$

$$\text{Area of } P = \frac{9}{16} \times 288$$

$$= 162 \text{ cm}^2$$

The area of the smaller triangle is 162 cm^2.

EXERCISE 13.08 ANSWERS ON P. 666

Areas of similar figures U F R

EXAMPLE 19

1 2 similar rectangles have sides in the ratio 1 : 4. Find the ratio of their areas.

2 2 similar circles have radii in the ratio 1 : 2.5. Find the ratio of their areas.

3 The ratio of the matching sides for pairs of similar figures are as shown.
Find the ratio of the areas for each pair.

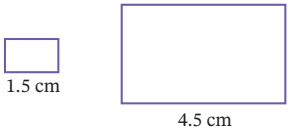
a 10 : 1 **b** 5.5 : 1 **c** 3 : 4 **d** 5 : 2

e 4 : 7 **f** 1 : 10 **g** 8 : 5 **h** 2 : 9

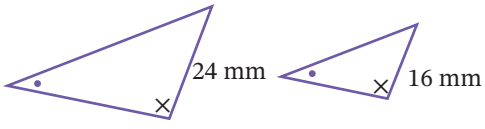
EXAMPLE 20

4 Find the ratio of the areas for each pair of similar figures.

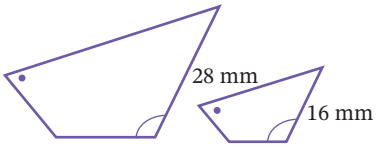
a



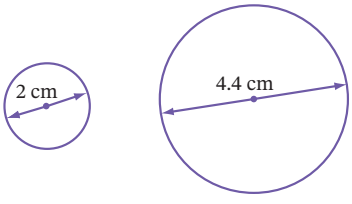
b



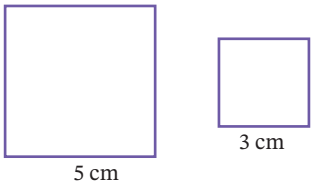
c



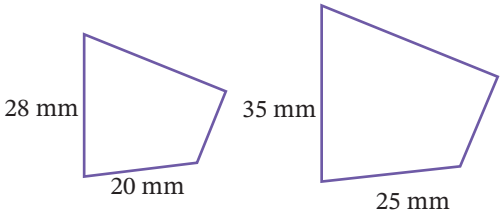
d



e



f



Foundation | Standard | Complex

PATH

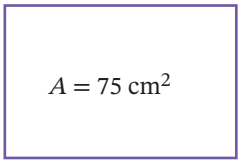
EXAMPLE 21

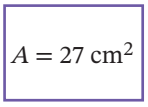
5 For each ratio between the areas of a pair of similar figures, find the ratio of the lengths of the matching sides.

a 64 : 49 **b** 36 : 1 **c** 9 : 400 **d** 1 : 20.25

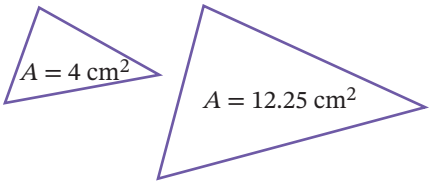
6 Find the ratio of the length of matching sides for each pair of similar figures.

a

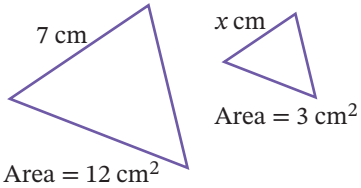




b



7 Find the value of x if these 2 triangles are similar.



EXAMPLE 22

8 2 similar triangles have matching sides in the ratio 4 : 1. If the area of the smaller triangle is 192 cm^2, find the area of the larger triangle. R

9 2 squares have sides in the ratio 3 : 4. If the area of the larger square is 288 cm^2, find the area of the smaller square. R

10 2 similar quadrilaterals have matching sides in the ratio 8 : 5. If the area of the smaller quadrilateral is 400 mm^2, find the area of the larger quadrilateral. R

11 2 circles have radii in the ratio 3 : 7. Find the area of the smaller circle if the area of the larger circle is 154 m^2. R

12 2 similar rectangles have areas in the ratio 36 : 121. If the width of the smaller rectangle is 84 mm, find the width of the larger rectangle. R

13 2 similar triangles have areas in the ratio 4 : 9. If the length of the base of the smaller triangle is 15 mm, find the length of the base of the larger triangle. R

14 2 squares have areas in the ratio 49 : 25. The length of the larger square is 17.5 cm. Find the length of the smaller square. R

15 If the radius of a circle is tripled, how has its area changed? R

16 If the area of a square is reduced by a factor of 4, how have the sides changed? R

17 The area of an equilateral triangle is decreased to $\frac{9}{16}$ of its original amount. How does its new area compare to its original area? R

18 A rectangle is reduced using a scale factor of $\frac{1}{2}$. If the area of the rectangle was 200 cm^2, what is the area of the reduced rectangle? R

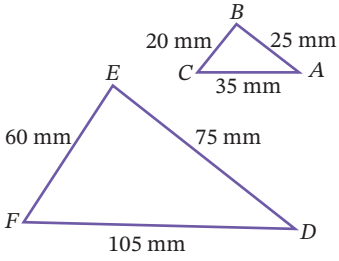
Investigation

Similar triangles

Work in groups of 3 or 4 to complete this investigation.

You will need: paper, a ruler, a pencil, a protractor, compasses and scissors.

1 **a** Draw $\triangle ABC$ with $AB = 25$ mm, $AC = 35$ mm and $BC = 20$ mm.



b How many different triangles can be drawn with sides of 25 mm, 35 mm and 20 mm? Is this triangle unique?

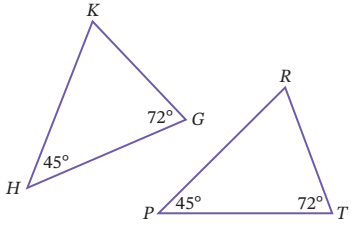
c Draw $\triangle DEF$ with $DE = 75$ mm, $DF = 105$ mm and $EF = 60$ mm.

d Are the matching sides of the triangles in the same ratio?

e Use your protractor to measure the angles of both triangles. Are matching angles equal?

f Is $\triangle ABC$ similar to $\triangle DEF$?

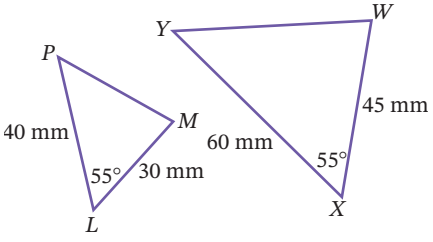
2 **a** Draw $\triangle GHK$ with $\angle G = 72°$ and $\angle H = 45°$.



b Draw $\triangle TPR$ with $\angle T = 72°$ and $\angle P = 45°$. Is $\triangle GHK \sim \triangle TPR$?

c Compare your triangles with those of other students. Are they all similar?

3 **a** Draw $\triangle LMP$ with $LM = 30$ mm, $LP = 40$ mm and $\angle L = 55°$ (the included angle).

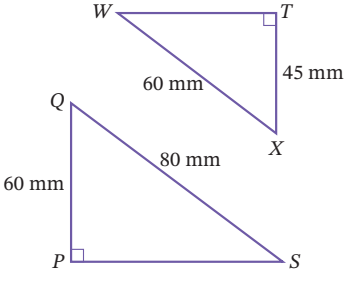


b Draw $\triangle XWY$ with $XW = 45$ mm, $XY = 60$ mm and $\angle X = 55°$.

c Find the ratio of 2 pairs of matching sides $\frac{LM}{XW}$ and $\frac{LP}{XY}$. Are they equal?

d Are the 2 triangles similar? Explain.

4 **a** Construct $\triangle PQS$ with $PQ = 60$ mm, $\angle P = 90°$ and $QS = 80$ mm (the hypotenuse).



b Construct $\triangle TXW$ with $TX = 45$ mm, $\angle T = 90°$ and $XW = 60$ mm.

c Calculate the ratios of matching sides $\frac{PQ}{TX}$ and $\frac{QS}{XW}$.

d Are the triangles similar? Explain.

5 **a** Compare and discuss your results with other students.

b What tests are necessary for 2 triangles to be similar?

c How do these conditions compare to the conditions for congruent triangles?

Foundation | Standard | Complex

13.09 Tests for similar triangles

PATH

Video
Similar triangles

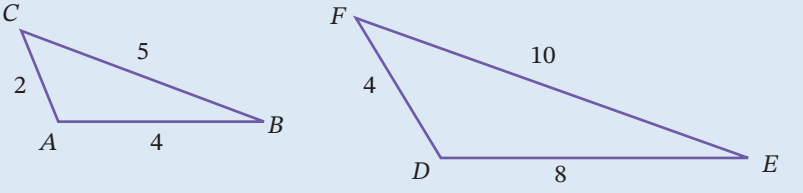
There are 4 sets of conditions that can be used to determine if 2 triangles are similar. These are called the **tests for similar triangles** or **similarity tests**.

Similarity tests

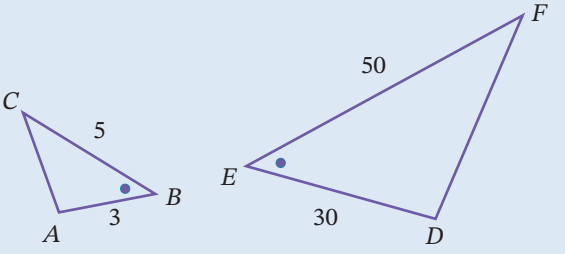
There are 4 tests for similar triangles.

Two triangles are similar if:

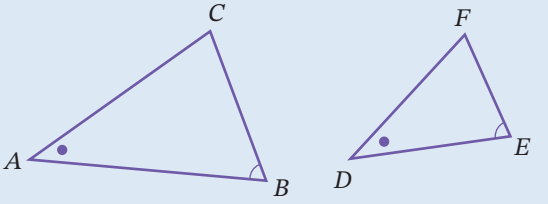
- the 3 sides of one triangle are proportional to the 3 sides of the other triangle **('SSS')**



- 2 sides of one triangle are proportional to 2 sides of the other triangle, and the **included** angles are equal **('SAS')**

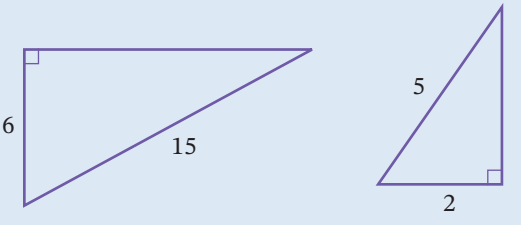


- 2 angles of one triangle are equal to 2 angles of the other triangle **('AA' or 'equiangular')**



Equiangular means 'equal angles'

- they are right-angled and the hypotenuse and a second side of one triangle are proportional to the hypotenuse and a second side of the other triangle **('RHS').**

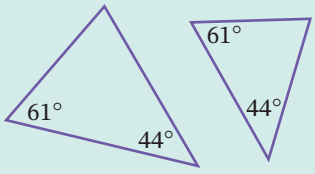


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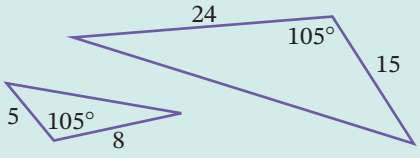
Example 23

Which test can be used to prove that each pair of triangles are similar?

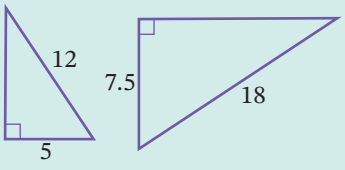
a



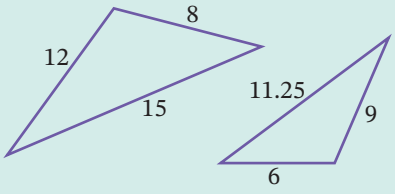
b



c



d



Solution

a 2 pairs of angles are equal, or equiangular ('AA').

b 2 pairs of matching sides are in the same ratio $\frac{15}{5} = \frac{24}{8} = 3$ and the included angles in both triangles are 105° ('SAS').

c In both right-angled triangles, the pairs of hypotenuses and second sides are in the same ratio $\frac{7.5}{5} = \frac{18}{12} = \frac{3}{2}$ ('RHS').

d All 3 pairs of matching sides are in the same ratio $\frac{11.25}{15} = \frac{9}{12} = \frac{6}{8} = \frac{3}{4}$ ('SSS').

PATH

Video
Tests for similar triangles

13.09

PATH

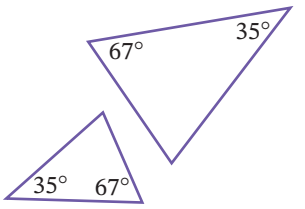
EXERCISE 13.09 ANSWERS ON P. 666

Tests for similar triangles U F R C

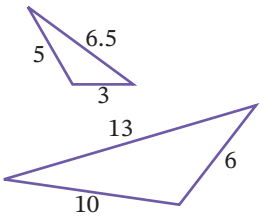
EXAMPLE 23

1 Which test can be used to prove that each pair of triangles are similar? R C

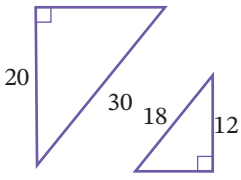
a



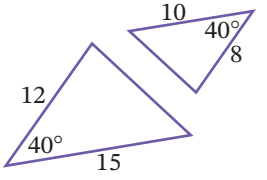
b



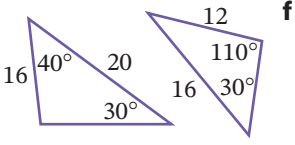
c



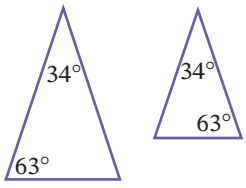
d



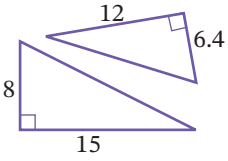
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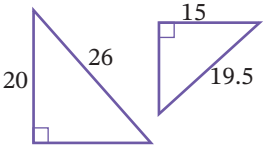
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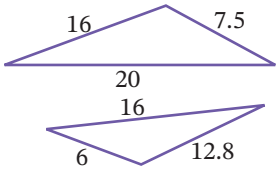
g



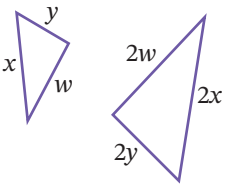
h



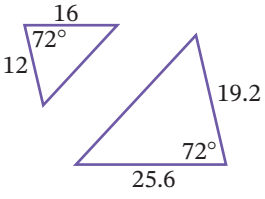
i



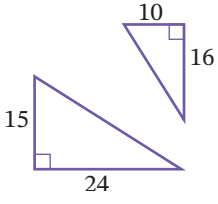
j



k

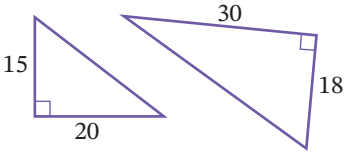


l

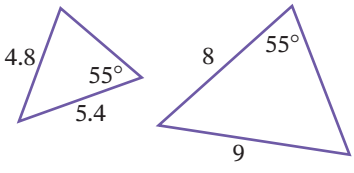


2 Which pair of triangles are similar? Select the correct answer **A**, **B**, **C** or **D**. R

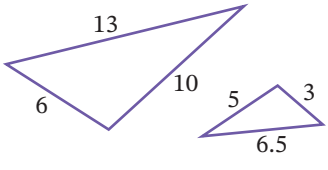
A



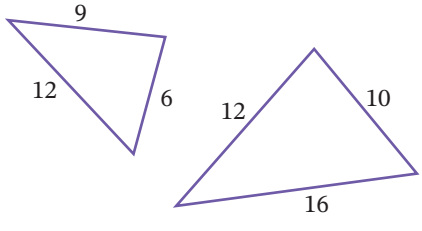
B



C



D

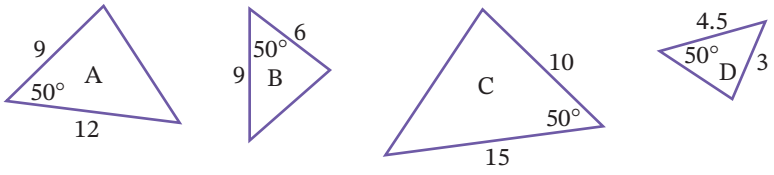


Foundation Standard Complex

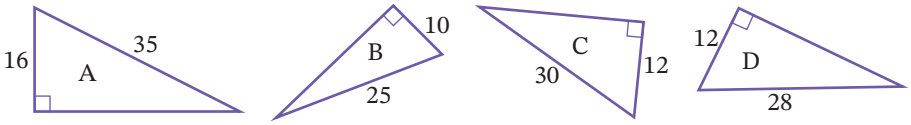
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3 For each set of triangles, find the pair of similar triangles. R

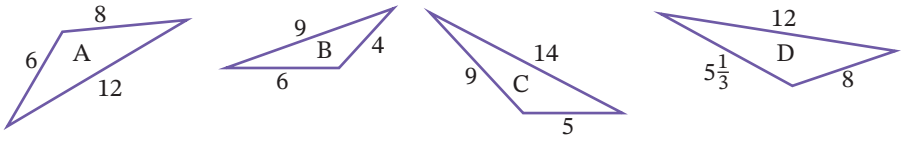
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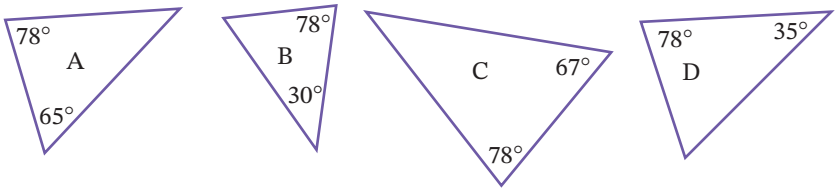
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c

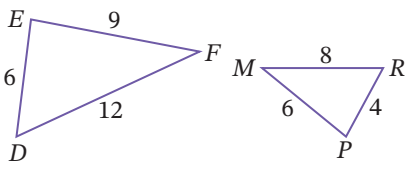


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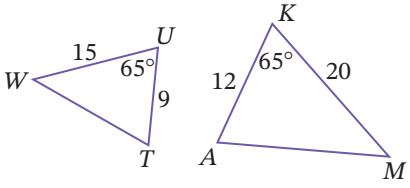


4 Use the correct notation to write a similarity statement relating each pair of similar triangles. R C

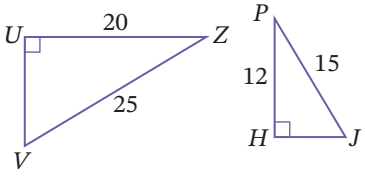
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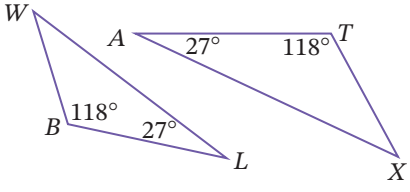
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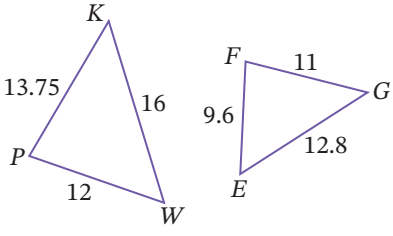
c



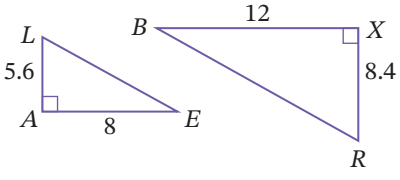
d



e



f



5 Similar triangles have 3 pairs of matching angles. So why is the similarity test AA and not AAA? R C

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13.10 Similar triangle proofs

PATH

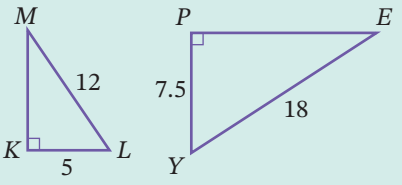
To formally prove that 2 triangles are similar, we use a specific format that involves applying one of the 4 similarity tests SSS, SAS, AAA or RHS.

Worksheets
Congruent and similar triangle proofs
Complete the proofs

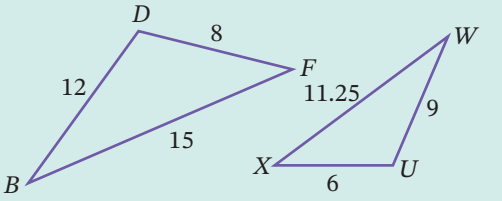
Example 24

Prove that each pair of triangles are similar.

a



b



Solution

a In $\triangle MKL$ and $\triangle EPY$:

$\frac{ML}{EY} = \frac{12}{18} = \frac{2}{3}$

$\frac{KL}{PY} = \frac{5}{7.5} = \frac{2}{3}$

$\therefore \frac{ML}{EY} = \frac{KL}{PY}$

$\angle K = \angle P = 90°$

$\therefore \triangle MKL \sim \triangle EPY$

(hypotenuses and another pair of matching sides of right-angled triangles in proportion, or 'RHS')

b In $\triangle DFB$ and $\triangle UXW$:

$\frac{BF}{WX} = \frac{15}{11.25} = \frac{4}{3}$

$\frac{BD}{WU} = \frac{12}{9} = \frac{4}{3}$

$\frac{DF}{UX} = \frac{8}{6} = \frac{4}{3}$

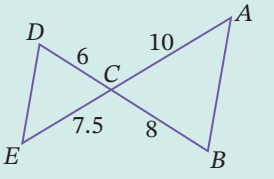
$\therefore \frac{BF}{WX} = \frac{BD}{WU} = \frac{DF}{UX}$

$\therefore \triangle DFB \sim \triangle UXW$

(3 pairs of matching sides in proportion, or 'SSS')

Example 25

Prove that $\triangle ABC \sim \triangle EDC$.



Solution

In $\triangle ABC$ and $\triangle EDC$:

$\frac{AC}{EC} = \frac{10}{7.5} = \frac{4}{3}$ $\quad$ $\frac{BC}{DC} = \frac{8}{6} = \frac{4}{3}$

$\therefore \frac{AC}{EC} = \frac{BC}{DC}$

$\angle ACB = \angle DCE$ (vertically opposite angles)

$\therefore \triangle ABC \sim \triangle EDC$ (2 pairs of matching sides in proportion and the included angles equal, or 'SAS')

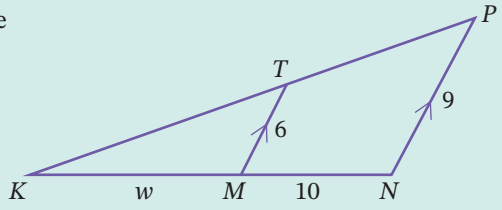
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PATH

13.10

Example 26

Prove that $\triangle KMT \sim \triangle KNP$ and hence find the value of w.



Solution

In $\triangle KMT$ and $\triangle KNP$:

$\angle K$ is common.

$\angle KMT = \angle KNP$ (corresponding angles, $MT \parallel NP$)

$\therefore \triangle KMT \sim \triangle KNP$ (equiangular, or 'AA')

$\therefore \dfrac{w}{w+10} = \dfrac{6}{9}$ (matching sides of similar triangles)

$$\begin{aligned} 9w &= 6(w+10) \\ &= 6w + 60 \\ 3w &= 60 \\ w &= 20 \end{aligned}$$

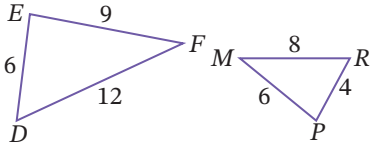
EXERCISE 13.10 ANSWERS ON P. 666

Similar triangle proofs UFRC

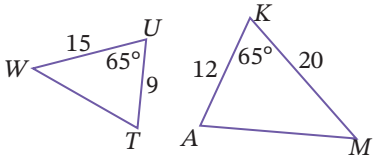
EXAMPLE 24

1 Prove that each pair of triangles are similar. R C

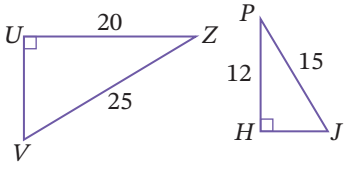
a



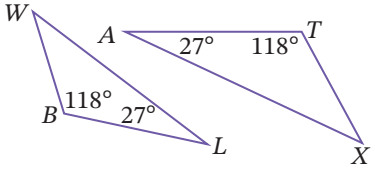
b



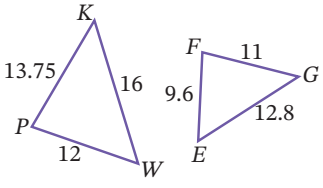
c



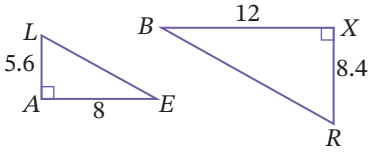
d



e



f



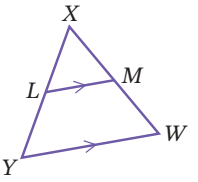
Foundation Standard Complex

PATH

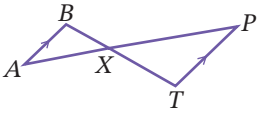
EXAMPLE 25

2 Prove that each pair of triangles are similar. R C

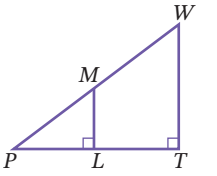
a $\triangle XYW \sim \triangle XLM$.



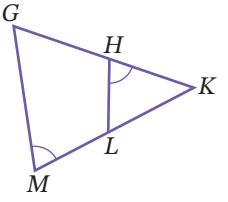
b $\triangle ABX \sim \triangle PTX$.



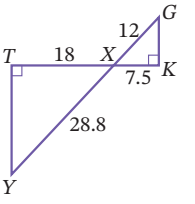
c $\triangle PLM \sim \triangle PTW$.



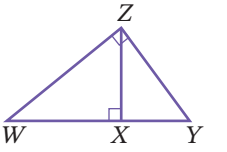
d $\triangle GKM \sim \triangle LKH$.



e $\triangle YTX \sim \triangle GKX$



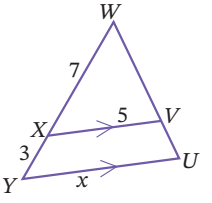
f $\triangle WZY \sim \triangle WXZ$



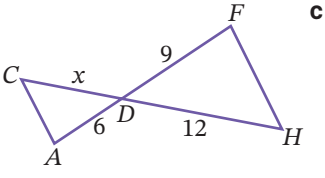
3 Find the value of x in each diagram after proving that 2 triangles are similar. R C

EXAMPLE 26

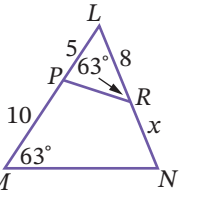
a



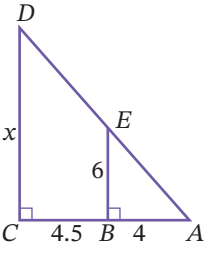
b



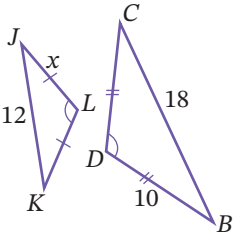
c



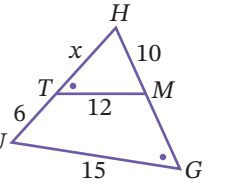
d



e

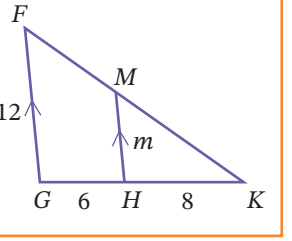


f



4 **a** Prove $\triangle KGF \sim \triangle KHM$. R C

b Find the value of m.



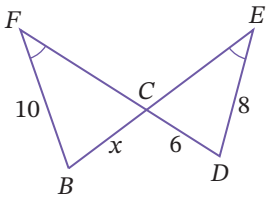
Foundation Standard Complex

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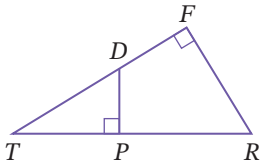
13.10

5 Prove $\triangle BCF \sim \triangle DCE$ and hence find the value of x. R C



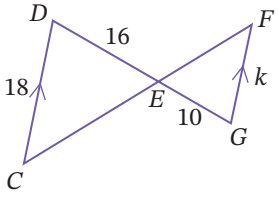
6 **a** Prove that $\triangle TFR \sim \triangle TPD$. R C

b If $FR = 9$ cm, $DP = 5$ cm and $TD = 8$ cm, find the length of TR.

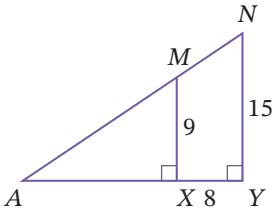


7 **a** Prove that $\triangle CDE \sim \triangle FGE$. R C

b Find the value of k.

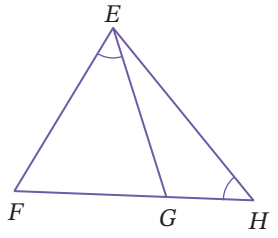


8 Prove $\triangle AXM \sim \triangle AYN$ and hence find the length of AX. R C



9 **a** Prove that $\triangle FGE \sim \triangle EFH$. R C

b If $FG = 16$ cm and $EF = 20$ cm, find the length of FH.



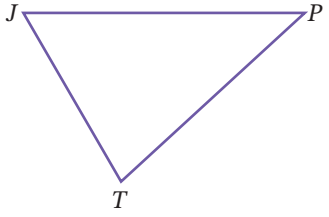
Foundation Standard Complex

Worksheet
Investigating paper sizes

Power plus ANSWERS ON P. 666

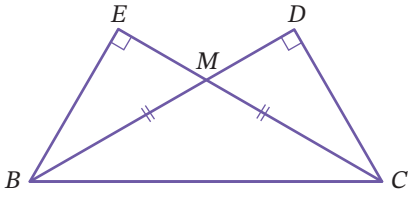
1 Construct $\triangle JPT$ by:

- **a** using compasses and the lengths of the 3 sides
- **b** using the length of one side and the size of the 2 angles
- **c** using the length of 2 sides and the size of one angle.



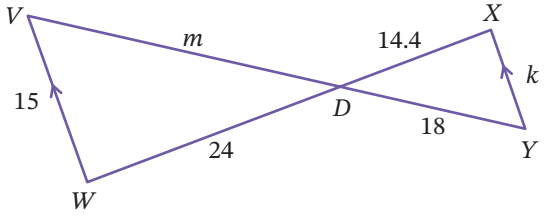
2 **a** List the information given in this diagram.

- **b** Why is $\angle BME = \angle CMD$?
- **c** Show that $\triangle BME \equiv \triangle CMD$.
- **d** Why is $EM = DM$?
- **e** Why is $\angle EBM = \angle DCM$?

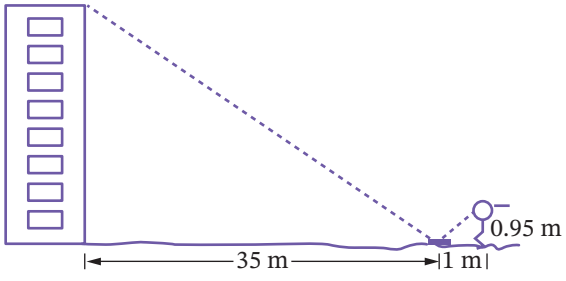


3 **a** What information is shown in the diagram?

- **b** Why is $\angle V = \angle Y$ and $\angle W = \angle X$?
- **c** Show that $\triangle VWD \sim \triangle YXD$.
- **d** Find the values of k and m.



4 Alexa places a mirror on the ground. She stands 1 m from the mirror and then bobs down until she can see the top of a building in the mirror. Her eyes are 95 cm above ground level when she can see the building. If the mirror is 35 m from the base of the building, what is the height of the building?



5 Is each statement true or false? Give reasons.

- **a** All equilateral triangles are similar.
- **b** All isosceles triangles are similar.
- **c** All circles are similar.
- **d** All rectangles are similar.
- **e** All squares are similar.
- **f** All regular pentagons are similar.

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CHAPTER 13 REVIEW

Language of maths

AAS	bisect	congruence test	congruent (≡)
diagonal	enlargement	hypotenuse	image
included angle	matching	original	parallel
proportional	ratio	reduction	RHS
SAS	scale	scale drawing	scale factor
similar (~)	similarity test	SSS	vertex/vertices

1 When an original figure is enlarged or reduced, what is the new figure called?

2 What is the angle between 2 given sides of a triangle called?

3 What is the symbol and meaning of 'is congruent to'?

4 What is the symbol and meaning of 'is similar to'?

5 Which word from the above list is another name for 'corresponding'?

6 What happens to a shape that is changed by a scale factor of 3?

7 What are the 4 tests for congruent triangles?

8 What does RHS stand for:

a in congruent triangles? **b** in solving equations?

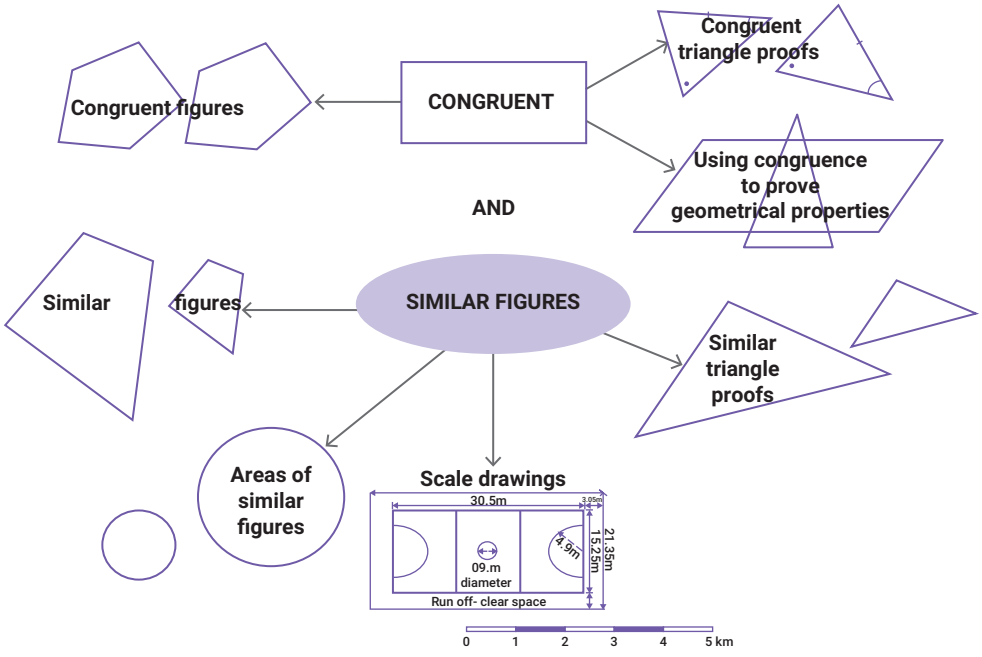
Quiz
Language of maths

Puzzle
Congruence and similarity crossword

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Similar figures



Quiz
Test yourself 13

TEST YOURSELF 13

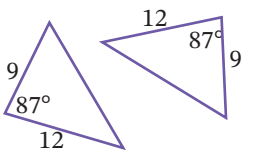
ANSWERS ON P. 667

PATH

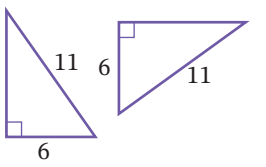
1 Which congruency test (SSS, SAS, AAS or RHS) can be used to prove that each pair of triangles are congruent?

13.02

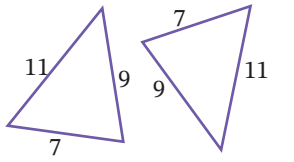
a



b

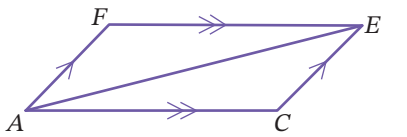


c



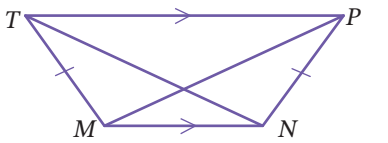
13.03

2 If $ACEF$ is a parallelogram, prove that $\triangle ACE \equiv \triangle EFA$.



13.03

3 If $MN \parallel TP$, $\angle MNP = \angle NMT$ and $TM = PN$, prove that $\triangle MNP \equiv \triangle NMT$.

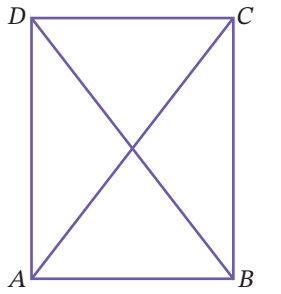


13.04

4 $ABCD$ is a rectangle.

a Prove that $\triangle ABD \equiv \triangle BAC$.

b Hence prove that the diagonals of a rectangle are equal in length.



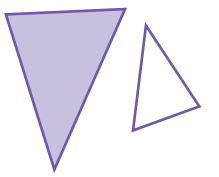
13.05

5 By measurement, find the scale factor for each pair of similar figures.

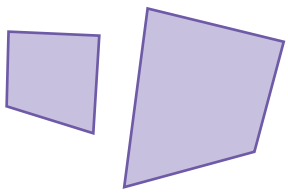
a



b



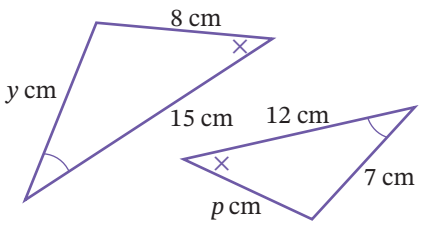
c



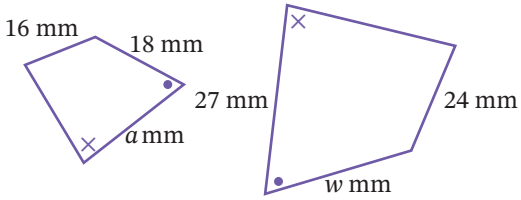
13.06

6 Find the value of each variable in each pair of similar figures.

a



b



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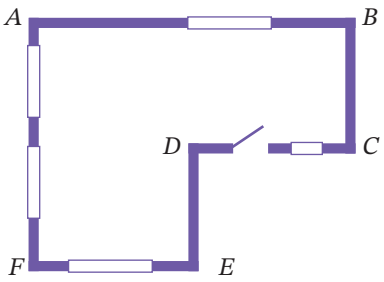
PATH

13.07

7 This floor plan of a holiday house is drawn to a scale of 1: 200. What is the actual (internal) length of:

a AB?

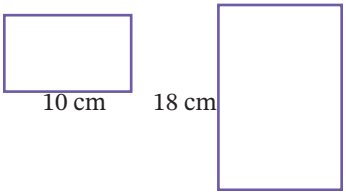
b CD?



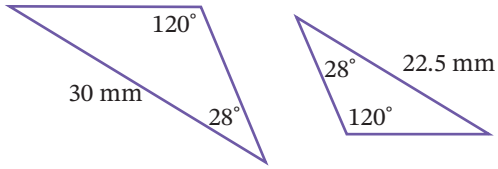
13.08

8 Find the ratio of the areas for each pair of similar figures.

a



b



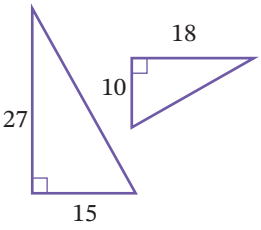
13.08

9 Two similar trapeziums have areas in the ratio 25 : 4. If the perpendicular height of the smaller trapezium is 40 cm, find the perpendicular height of the larger trapezium.

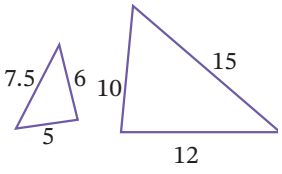
13.09

10 Which test can be used to prove that each pair of triangles is similar?

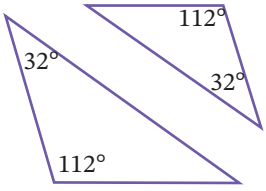
a



b



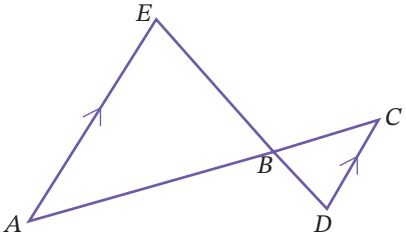
c



13.10

11 **a** Prove that $\triangle ABE \sim \triangle CBD$.

b If $DC = 8$ cm, $AE = 14$ cm and $CB = 7$ cm, find the length of AB.



PRACTICE SET 4 ANSWERS ON P. 667

11.04

1 Test whether the point (2, 3) lies on each line.

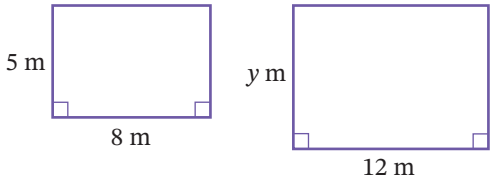
a $2x + y = 4$ **b** $y = 2x - 1$ **c** $x + y = 5$

13.06

2 The rectangle on the left has been enlarged to make the rectangle on the right.

a What is the scale factor?

b Find the value of y.



12.01

3 The probability of Eva hitting a bullseye on a target is 0.2. Find the probability of Eva missing the bullseye.

PATH

12.04

4 A class was surveyed on whether its students could swim. The results are shown.

	Can swim	Cannot swim
Boys	13	2
Girls	9	3

a Find the probability of randomly selecting a student who is:

i a boy **ii** a boy or someone who cannot swim

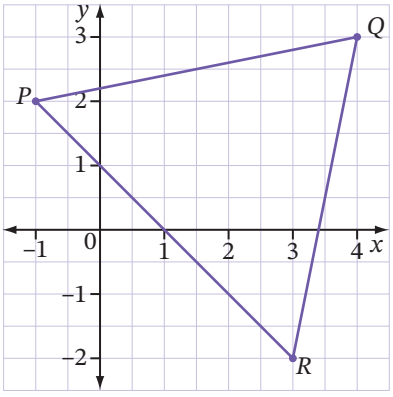
iii a girl who cannot swim **iv** a girl or someone who can swim, but not both

b If a swimmer is selected at random, what is the probability that it is a boy?

13.07

5 The scale on a tourist map of Sydney is 2 cm = 500 m. If the scaled distance from Circular Quay station to the Opera House is 2.5 cm, what is the actual distance?

Questions **6** to **9** refer to this triangle that has vertices $P(-1, 2)$, $Q(4, 3)$ and $R(3, -2)$.



11.01

6 Find the length of:

a side PQ, as a surd

b side PR, correct to one decimal place

11.01

7 Show that $\triangle PQR$ is isosceles.

11.02

8 Find the coordinates of the midpoint of side PR.

11.03

9 Find the gradient of:

a PQ

b PR

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10 Graph $y = 4 - 2x$ on a number plane and write: 11.04

a its x-intercept

b its y-intercept.

11 Daniel rolled a biased die 90 times and the number 6 came up 24 times. 12.02

a What is the observed probability of rolling a 6?

b If the same die was rolled 400 times, what is the expected frequency of rolling a 6?

12 **a** Sketch the graph of $y = x^2 - 2$. 11.10

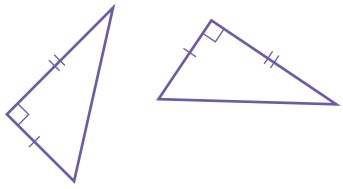
b Write the coordinates of the vertex.

c Is the graph concave up or concave down?

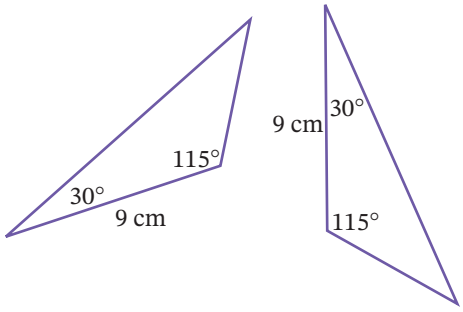
PATH

13 Which congruence test (SSS, SAS, AAS or RHS) can be used to prove that each pair of triangles are congruent? 13.02

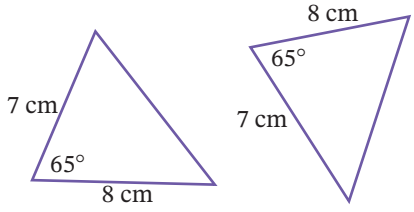
a



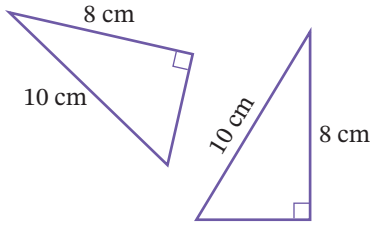
b



c



d



14 What is the equation of the horizontal line passing through (–2, 4)? 11.04

15 The numbers 1 to 30 are written on cards and the cards are placed in a bag. If one card is drawn at random from the bag, what is the probability that it is: 12.01

a a multiple of 5?

b greater than 10?

16 **a** Find the gradient and y-intercept of the line with equation $y = \frac{1}{2}x + 3$. 11.05

b Graph a line with gradient 2 and y-intercept –5 and write its equation.

17 The increase in pressure experienced by a scuba diver is directly proportional to the depth under the water. The increase in pressure at depth 40 m is 235 kPa (kilopascals). PATH 11.09

a What increase in pressure is experienced at a depth of 32 m?

b At what depth, to the nearest metre, is the increase in pressure 735 kPa?

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12.05

18 2 dice are rolled together. Copy and complete the table to show the sample space.

		1st die					
		1	2	3	4	5	6
2nd die	1						
	2					5, 2	
	3						
	4		2, 4				
	5						
	6						

Find the probability of rolling:

a a 3 and a 6 (in any order)

b at least one 2

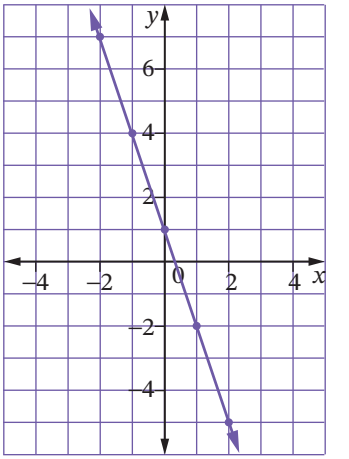
c 2 numbers the same

d a 4 or a 5 but not both

11.06

19 **a** Find the gradient of this line.

b Find the equation of this line.

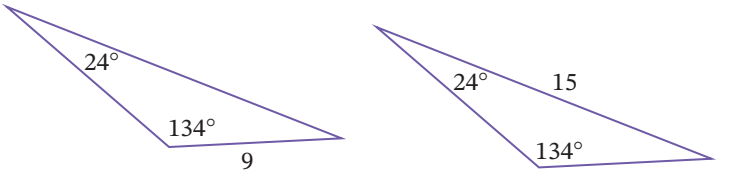


11.13

20 What is the equation of a circle with centre (0, 0) and radius 4 units?

13.09

21 Which test can be used to prove that this pair of triangles are similar?



11.13

22 Match each equation to the correct description of its graph.

a $y = x^2 + 9$

b $y = x + 9$

c $y = 9$

d $x = 9$

e $x^2 + y^2 = 9$

A straight line

B circle

C parabola

D vertical line

E horizontal line

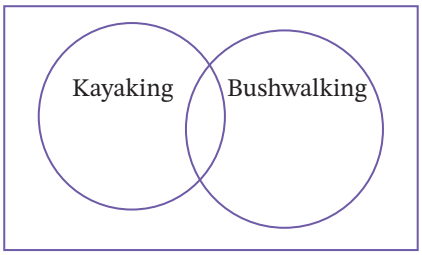
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PATH

12.03

23 40 students at a school camp could select kayaking or bushwalking as an activity, or both. A total of 35 students chose kayaking, 20 chose bushwalking, including some who chose both. There weren't any students who chose neither.



a Copy and complete the Venn diagram representing these students.

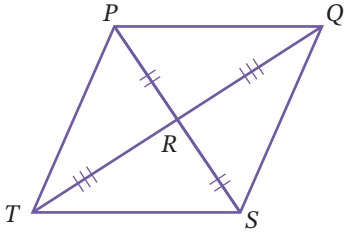
b What is the probability that a student selected at random:

i chose kayaking and bushwalking?

ii chose kayaking or bushwalking but not both?

13.04

24 The diagonals of this rhombus divide the rhombus into 4 congruent triangles.



a Prove that $\triangle PRQ \equiv \triangle PRT$.

b Hence find the size of $\angle PRQ$.

c Copy and complete: The diagonals of a rhombus _____ each other at _____ _____.

25 Two similar rectangles have their areas in the ratio 1 : 7.84. If the length of the larger rectangle is 5.04 m, calculate the length of the smaller rectangle.

13.08

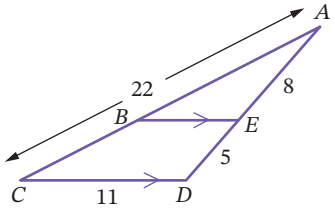
13.10

26 **a** Prove that $\triangle ABE \sim \triangle ACD$.

b Find the length of:

i BE

ii BC



11.12

27 A ball is thrown into the air and its height can be modelled by the equation $h = 3x - 0.5x^2$, where h is the height above the ground after x seconds.

a Copy and complete this table of values and then graph the equation.

Time, x (seconds)	0	1	2	3	4	5	6
Height, h (metres)							

b What is the maximum height the ball reaches?

c What is the height of the ball after 1.5 seconds (correct to 1 decimal place).

d At what times is the ball at a height of 3 m?

e How far does the ball travel in:

i the third second?

ii the fifth second?